Engine Model(s): PT6A

70-00-00

STANDARD PRACTICES - MAINTENANCE PRACTICES

1. General

A. During disassembly, tag and mark all parts, clips, and brackets as to their location and position to prevent incorrect installation.

B. During removal of tubes or engine parts look for indications of scoring, burning or other undesirable conditions. Tag unserviceable parts and units for investigation and possible repair.

C. Extreme care shall be taken to prevent dust, dirt, lockwire, nuts, washers or other foreign matter from entering the engine. It cannot be overemphasized that this precaution applies whenever work is done on the engine either on the wing or off the wing. Suitable plugs, caps, and other covering shall be used to protect all openings as they are exposed. Dust caps used to protect open tubes against contamination shall always be installed over tube ends and not in the tube. Flow through the lines may be blocked if lines are inadvertently installed with dust caps in the tube ends.

D. If at any time items are dropped into the engine, the assembly process must stop until the dropped articles are located, even though this may require a considerable amount of time and labor. Before assembling or installing any part, be sure it is thoroughly clean.

E. Lockwire, lockwashers, tablocks, tabwashers, or cotterpins must never be reused. All lockwire and cotterpins must fit snugly in holes drilled in studs and bolts for locking purposes. Install a cotterpin so that the head fits into the castellation of the nut, and unless otherwise specified, bend one end of the pin back over the stud or bolt and the other end down flat against the nut. Only lockwire and cotterpins made of corrosion resistant steel shall be used. Bushing plugs shall be lockwired to the assembly boss or case; do not lockwire the plug to the bushing.

F. Replace all gaskets, packings, and rubber parts at reassembly. Make sure that new nonmetallic parts to be installed (such as an oil seal) show no sign of having deteriorated in storage.

G. To protect critical areas of engine parts (such as compressor and turbine disks) against scratches and nicks, tool surfaces contacting these areas must be covered with protective material.

H. During replacement of components, note condition of any removed preformed packings. Pieces accidentally torn or cut from packings must be retrieved, regardless of size, even if this entails a systematic disassembly and cleaning of the system. Make sure that new packings are free of cuts, flashings and deformities which may be sheared off at installation.

I. When installing engine parts that require the use of a hammer to facilitate assembly or installation, use only a plastic or rawhide hammer.

J. Whenever pressure sensitive tape has been applied to any part, the tape must be removed and the part thoroughly cleaned of all tape residue prior to being subjected to high temperature during engine run. Test results indicate that all tapes are capable of causing surface attack and/or reduction in tensile ductility as temperature increases. Do not leave tape or tape residue

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on engine parts.

K. If any part has been coated with corrosion preventive compounds, all traces of the compound and accumulated foreign matter must be removed.

2. Safe Handling of Engine Parts with Suspected Hexavalent Chromium Residue

A. General

(1) This procedure gives the safety instructions that are applicable for assemblies and components that may contain hexavalent chromium (Cr(VI)) residue.

(2) Hexavalent Chromium (Cr(VI)) compounds may form on certain engine hot section and other parts as a result of interactions at engine operating temperatures with certain dry film lubricants or assembly consumables, atmospheric dust (e.g., calcium oxide and/or calcium hydroxide) drawn into the engine during operation, and chromium at the surface of nickel-based and stainless-steel alloy parts. The formation of Cr(VI) may be most prevalent on parts that operate in the temperature range 750 °F to 1800 °F (399 °C - 982 °C). Even if residue is not visible, Cr(VI) residue may be present on the surface of parts operating at those temperatures. Visible residue may range from pale white-beige to yellow, orange, light brown and/or dark brown. Hexavalent Chromium (Cr(VI)) may also appear as a powder or staining.

(3) Exposure to high levels of airborne, oral or dermal Cr(VI) compounds can be associated with an increased health hazard, and should be avoided. For additional information concerning potential adverse health effects associated with Cr(VI) exposure, refer to your local occupational/environmental health and safety agency, for example, https://www.osha.gov/hexavalent-chromium.

(4) Engine maintenance personnel or others who may come into contact with potentially-affected engine parts should comply with applicable governmental regulatory guidelines to minimize possible worker exposure to Cr(VI) residue.

B. General Work Practices

WARNING: EXPOSURE TO HIGH LEVELS OF AIRBORNE, ORAL OR DERMAL HEXAVALENT CHROMIUM COMPOUNDS CAN BE ASSOCIATED WITH AN INCREASED HEALTH HAZARD, AND SHOULD BE AVOIDED. ENGINE MAINTENANCE PERSONNEL OR OTHERS WHO MAY COME INTO CONTACT WITH POTENTIALLY-AFFECTED ENGINE PARTS SHOULD COMPLY WITH APPLICABLE GOVERNMENTAL REGULATORY GUIDELINES TO MINIMIZE POSSIBLE WORKER EXPOSURE TO HEXAVALENT CHROMIUM RESIDUE.

(1) Avoid generating airborne dust and avoid skin and eye contact with any residue found on engine parts. Compressed air should not be used to remove residue or clean parts. Operations such as grinding, buffing or media blasting should be conducted after part cleaning and removal of residue, and only with appropriate controls in place to limit potential exposure to Cr(VI).

(2) If potentially-contaminated component parts are to be moved or transported, take precautions to minimize direct contact with or release of residue during handling and transportation.

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C. Personal Protection and Hygiene

(1) Wear disposable nitrile or neoprene gloves and eye/face protection whenever inspecting or handling hot section parts to minimize possible exposure to Cr(VI) residue. Follow good personal hygiene (hand washing) after glove removal.

(2) In case of contact with the residue and/or stain, wash any areas of your skin that have not been protected with personal protective equipment and contact your local Environmental Health Safety (EHS) representative.

(3) Avoid food and beverage consumption and smoking in work areas where Cr(VI) may be present.

(4) If your clothing has the potential to become contaminated, wear protective disposable coveralls.

D. Cr(VI) Determination of Residue

(1) Contact your local EHS and/or Industrial Hygiene (IH) representative for approved methods to determine if residue contains Cr(VI).

E. Handling and Removal of Suspected or Confirmed Cr(VI) Residue

(1) Contact your local EHS representative and/or IH representative for guidance and instruction on handling and/or removal of suspected or confirmed Cr(VI) residue.

F. Shipping Guidance

(1) Contact your local EHS, IH representative and/or shipping representative for guidance on shipping engine material with suspected or confirmed Cr(VI).

G. Managing Suspected Hexavalent Chromium-Contaminated Waste Packaging Guidance

(1) Contact your local EHS and/or IH representative for guidance on managing suspected or confirmed Cr(VI) contaminated waste.

H. Unwanted Material Disposal

(1) Contact your local EHS and/or IH representative for guidance on disposing waste that has or potentially came in contact with the Cr(VI) residue, such as rags, Q-tips/brushes, wastewater, gloves, and disposable coveralls. These should be treated and properly disposed of according to local waste regulations.

3. <u>Standard Torques</u>

A. General

(1) Torque values given in the manual are to be interpreted:

(a) Pound-inches at room temperature.

(b) Angle of turn, in degrees.

(c) Stretch values in inches.

(2) Unless otherwise specified, thread lubricant shall be engine oil and shall be applied to parts

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which are to be tightened.

(3) If the part to be tightened is hot, allow sufficient time for the part to reach temperature equilibrium with the surrounding area before final torquing.

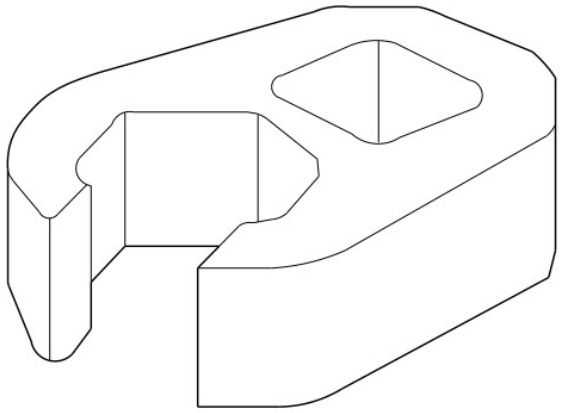
(4) Flange bolts shall be drawn up evenly. This can be achieved by having two mechanics working simultaneously 180 degrees apart.

(5) Tightening applications should be done slowly and evenly for consistency and the best possible accuracy. Check for adequate torque of self-locking fasteners before reuse. Discard fastener if locking capability is impaired. Do not repair self-locking fasteners.

(6) Six-point crowfoot adapters are recommended for applying torque values above 110 lbf.in. to tube fittings (Ref. Fig. 201).

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Figure 201 Six-Point Crowfoot Adapter



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(7) Self-locking nuts, self-locking bolts, and self-locking helical coil inserts shall be capable of meeting the torque requirements in Table 201.

Table 201 Torque Check for Reuse of Self-Locking Nuts, Bolts and Helical Coil Inserts (Pound-inches at Room Temperature and Lubricated With Engine Oil)

Mating Bolt Thread Size	Max Locking Torque	Min Breakaway Torque	Mating Bolt Thread Size	Max Locking Torque	Min Breakaway Torque
0.112-48NF-2A	3	0.5	0.112-40NC-2A	3	0.5
0.138-40NF-2A	6	1.0	0.125-40NC-2A	4	1.0
0.164-36NF-2A	9	1.5	0.138-32NC-2A	6	1.0
0.190-32UNF-3A	13	2.0	0.164-32NC-2A	9	1.5
0.250-28UNF-3A	30	3.5	0.190-24UNC-3A	13	2.0
0.3125-24UNF-3A	60	6.5	0.250-20UNC-3A	30	4.5
0.375-24UNF-3A	80	9.5	0.3125-18UNC-3A	60	7.5
0.4375-20UNF-3A	100	14.0	0.375-16UNC-3A	80	12.0
0.500-20UNF-3A	150	18.0	0.4375-14UNC-3A	100	16.5
0.5625-18UNF-3A	200	24.0	0.500-14UNC-3A	150	24.0
0.625-18UNF-3A	300	32.0	0.5625-12UNC-3A	200	30.0
0.750-16UNF-3A	400	50.0	0.625-11UNC-3A	300	40.0
0.875-14UNF-3A	600	70.0	0.750-10UNC-3A	400	60.0

NOTE: The breakaway torque specified is the minimum torque required to overcome static friction, and applies at installation or removal. The locking torque specified is the maximum torque required to overcome kinetic friction and applies at installation or removal.

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Mating Bolt Thread Size	Max Locking Torque	Min Breakaway Torque	Mating Bolt Thread Size	Max Locking Torque	Min Breakaway Torque

NOTE: Use the listed bolt sizes for torque testing cadmium plated nuts, carbon steel nuts, alloy steel nuts, and aluminum nuts.

4. Torque Indicating Devices

A. General

(1) Check torque indicating devices before using, and calibrate them by means of weights and a measured lever arm to make sure that there are no inaccuracies. Checking one torque wrench against another is not sufficient. Some wrenches are quite sensitive to the way they are supported during a tightening operation, and every effort must be made to adhere to the instructions furnished by the respective manufacturers.

B. Torque Wrench and Extensions

(1) Occasionally, it is necessary to use a special extension, or adapter wrench together with a standard torque wrench (Ref. Fig. 202). In order to arrive at the resultant required torque limits, the following formula shall be used:

T - Desired torque on the part.

E - Effective length of special extension or adapter.

L - Effective length of torque wrench.

A - Distance through which force is applied to part.

R - Reading on scale or dial of torque wrench.

$$R = \frac{LT}{A} = \frac{LT}{L + E}$$

Example:

A torque of 1440 lb.in. is desired on a part, using special extension having a length of three inches from center to center of its holes, and a torque wrench measuring 15 inches from center of handle or handle swivel pin to center of its square adapter. Then:

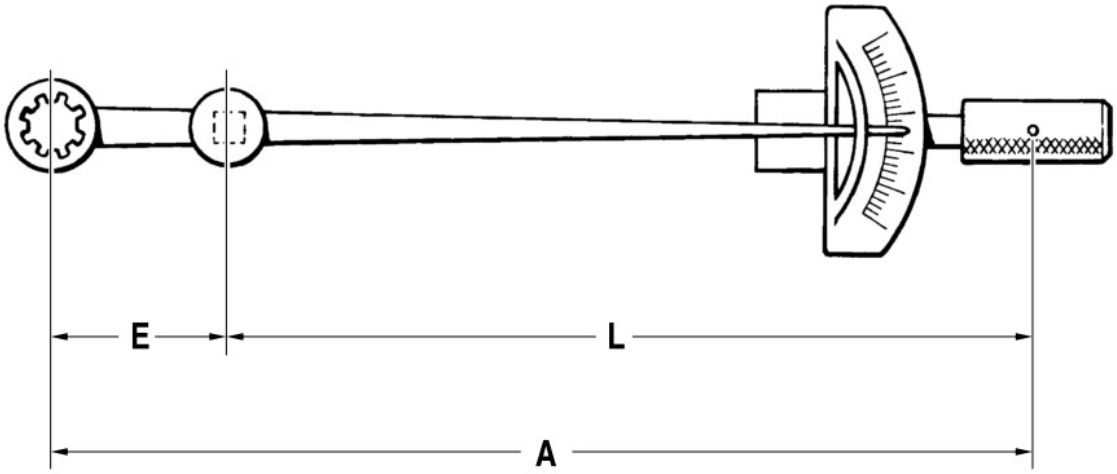
$$R = \frac{LT}{L + E} = \frac{15 \times 1440}{15 + 3} = 1200 \text{ lb.in.}$$

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(2) With the axis of the extension or adapter and the torque wrench in a straight line, tightening to a wrench reading of 1200 lb.in. will provide the desired torque of 1440 lb.in. on the part.

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Figure 202 Torque Wrench and Extension



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5. Torque Recommendations

A. Lubricated Parts

(1) All nut, bolt and screw torques should be obtained using a thread lubricant. The lubricant should be engine oil, or equivalent, unless otherwise specified. Torque requirements for interference fit application, such as studs and pipe plugs, may be obtained with or without lubrication, unless otherwise specified.

(2) Antigalling compound should be applied to all loose-fit spline drives which are external to the engine and have no other means of lubrication. For certain assembly procedures, molybdenum disulfide in either paste or powder form mixed with engine oil or grease may be used. Particular applications of molybdenum disulfide are indicated in the individual text portions of this manual. Make sure antiseize and antigalling compounds are applied in a thin even coat, and that excess is completely removed to avoid contamination of adjacent parts, passages or surfaces where the compound may cause malfunctioning, or failure, of engine.

(3) The torque values given within the text apply to nuts where the height of the nut is approximately equal to the major diameter of the thread. For jam nuts refer to REMOVAL/INSTALLATION. Torque values do not apply to hollow bolts and screws.

B. Nuts, Bolts and Screws

(1) Bolts and nuts on flanges with metal tubular gaskets must be initially torqued to the required torque and then retorqued until the torque value given in the relevant assembly instruction remains constant.

C. Self-locking Nuts

(1) The locking torque for self-locking nuts and helical coil inserts must be checked at assembly. The locking torque must not be less than the values shown (Ref. Table 202) unless otherwise specified.

(2) When checking self-locking torque, care must be taken to ensure the fastener is not seated. This ensures that only the torque necessary to overcome the friction holding the thread is measured.

NOTE: Self-locking nuts should be checked for effectivity of self-locking feature before reinstallation. Reject suspect nuts as necessary.

(3) For each item, note the torque necessary to turn nut on bolt before seating the nut.

(4) Add this torque to the value detailed in the assembly instructions for the application. Use this new value as the total applied torque.

Table 202 Locking Torque for Self-locking Nuts and Helical Coil Inserts

Thread Size	Torque lb.in. Min
0.112-40	0.5
0.112-48	0.5

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Thread Size	Torque lb.in. Min
0.138-32	1.0
0.164-32	1.5
0.164-36	1.5
0.190-24	2.0
0.190-32	2.0
0.2500-20	4.5
0.2500-28	3.5
0.3125-18	7.5
0.3125-24	6.5
0.3750-16	12.0
0.3750-24	9.5
0.4375-14	16.5
0.4375-20	14.0
0.5000-13	24.0
0.5000-20	18.0
0.5625-12	30.0
0.5625-18	24.0
0.6250-11	40.0
0.6250-18	32.0
0.750-10	60.0
0.750-16	50.0
0.875-9	82.0
0.875-14	70.0
1.000-8	110.0
1.000-12	90.0

D. Power Torque Wrench

(1) The retaining nut for the power turbines is removed and installed with a power torque

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wrench and appropriate adapter, supplied with the engine overhaul kit.

(2) The power torque wrench employs a high ratio geartrain operated by a handcrank (supplied with the tool) or a pneumatic ratchet drive tool. A $^{3}/_{4}$-inch square drive bar (also supplied with the tool) transmits the output torque of the geartrain to the socket wrench on the nut being turned. Two $^{1}/_{2}$-inch diameter pins on each space adapter transmit the counteracting load. Input and output shafts of the wrench rotate in the same direction.

(3) It must be emphasized that when using a countertorque adapter attached to the nut, the direction of the input drive is opposite to that normally required on the nut being turned, i.e., when tightening a nut, the input drive must be turned counterclockwise; when loosening, the reverse rotation applies.

(4) To tighten a nut, using the power torque wrench and appropriate countertorque adapter:

(a) Install socket wrench on nut.

(b) Install appropriate countertorque adapter on part being assembled.

NOTE: Observe instructions marked on adapter denoting direction of rotation required when torquing and untorquing.

(c) Install power torque wrench on adapter, and engage both pins.

(d) Insert 3/4-inch output drive bar into wrench output shaft, and turn until bar engages socket wrench.

(e) Turn ratchet (located on front face of power wrench) in same direction as that required to turn output driveshaft.

(f) Insert input drive source (handle or power-operated wrench) in input shaft, and turn in same direction as that required for output shaft.

(g) Continue turning until torque indicator reaches required reading on scale.

NOTE: Since the torque indicator is calibrated in pounds-feet, the appropriate conversion must be made when torquing to values specified in pounds-inches.

(h) To remove power wrench, reverse direction of rotation of input drive until indicator returns to zero (green band); this removes the holding pressure on the wrench, allowing it to be withdrawn from the adapter.

(i) Remove power wrench, adapter and socket wrench.

(5) To untorque, repeat the preceding steps (a) through (i), except for rotational direction of ratchet and input shaft, which must be reversed.

E. Lockwire and Cotterpin Requirements

(1) When tightening a castellated nut, alignment of the slot must be obtained without exceeding the maximum torque. If this is not possible, replace the nut with another one. After tightening the nut to the recommended torque, the nut must not be loosened to permit insertion of lockwire or cotterpin. If the slot in the nut or the lockwire hole in the bolt or screw is not correctly aligned at the minimum torque value given, the nut, screw or bolt should be further tightened to the next alignment position, but the maximum torque value

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given must not be exceeded. Should the alignment still be impossible without exceeding the maximum torque, back off the nut, screw or bolt one-half turn and retorque.

F. Slotted, Steel Locknuts (Prevailing Torque Type)

(1) Effective locking of slotted, steel locknuts on bolts or studs requires full engagement of all locknut threads. The chamfered section of the locknut ID does not exert force on the bolt or stud; therefore, it is not necessary that the bolt or stud be flush with, or protrude from, the outer face of the locknut.

G. Standard and Stepped Studs

(1) When the torque required to drive a stud to the correct protrusion does not reach the minimum value given, or exceeds the maximum value given, a new stud must be selected.

H. Hose, Tubing and Threaded Couplings

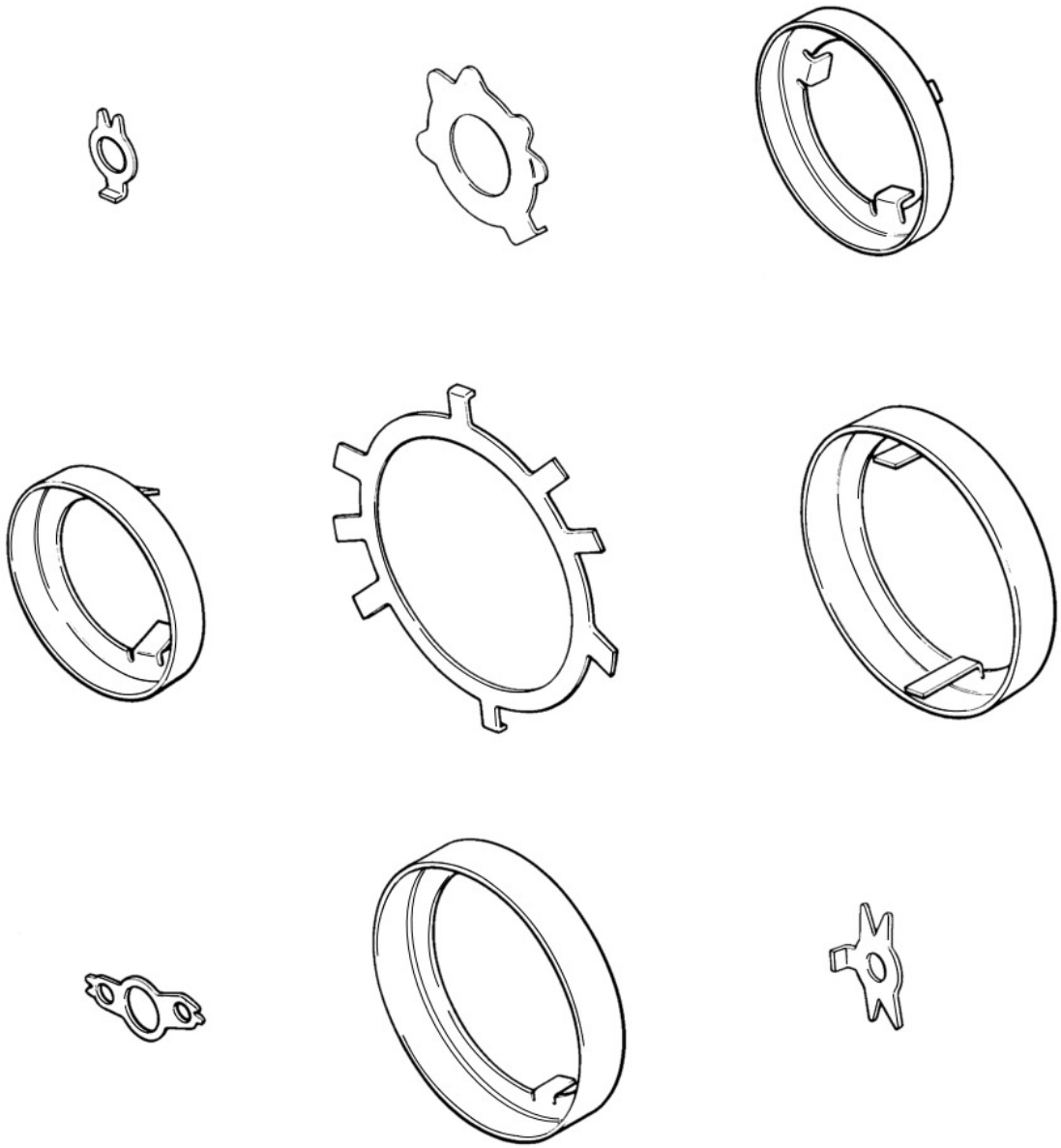
(1) If leakage occurs at a coupling, do not attempt to correct by overtorquing. Disassemble the fitting and check for nicks, burrs and/or foreign matter. Use new parts, as required, to rectify fault.

6. Locking Devices

A. Keywashers (Tab and Cup Type) (Ref. Fig. 203)

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Figure 203 Examples of Lockwashers



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CAUTION: KEYWASHERS ARE TO BE USED ONCE ONLY. INSTALL NEW KEYWASHER(S) AT EACH ASSEMBLY.

CAUTION: DO NOT USE SHARP EDGED TOOLS TO BEND OR SET KEYWASHER TABS. TAB DETACHMENT MAY OCCUR WITH SUBSEQUENT DAMAGE TO THE ENGINE.

(1) The terms keywasher, tabwasher and cupwasher are interchangeable, as used in this manual.

(2) When bending or setting the tabs on these washers, do not use sharp-pointed tools. Use of such tools can lead to subsequent failure of the locking tabs which, on becoming detached, can pass through the engine causing extensive damage.

(3) Key and tab washers typically have an anti turn tang that inserts into the engine part accepting the bolt. During torquing of bolt, it is important to make sure this tang is loaded in a counter clockwise direction within its hole so that the bolt and washer do not turn during operation.

(4) Assembly requirements for cupwashers:

(a) Examine the cupwasher for freedom from deep drawing score marks (especially in undercut adjacent to tangs) and for freedom from prior assembly/handling damage, i.e. bent tangs or prior stake marks.

(b) Examine contact surfaces for excessive roughness, burrs, or scores that may cause the washer to bend on the nut.

(c) Examine the nut for burrs, nicks or scratches on the face adjacent to the cupwasher that could cause the nut to pick up on the washer.

(d) The cupwasher must be lubricated on the nut side only, the opposite face must be clean and dry.

(e) With the cupwasher placed against the shaft slots opposite to direction of nut tightening, screw the nut fingertight.

(f) With an approved silver pencil, make an alignment mark on the cupwasher outside diameter and the component immediately adjacent to cupwasher.

(g) Tighten the nut in the normal manner.

(h) If the alignment mark has moved, remove and discard the cupwasher.

(i) Repeat the procedure with a new cupwasher.

(j) Stake the cupwasher.

B. Retaining Rings

CAUTION: THOROUGHLY INSPECT ALL RETAINING RINGS, INCLUDING SPIROLOX, FOR CONDITION. DISTORTED RINGS ARE NOT ACCEPTABLE AND MUST BE REPLACED.

(1) Retaining rings must be installed using approved retaining ring pliers. Internal type rings

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must not be compressed beyond the point where the ends of the ring meet. External type rings must be expanded just enough to allow installation without becoming bent. After installation, make sure that each ring is completely seated, without looseness or distortion in its groove.

C. Lockwire

(1) Except where otherwise specified, the wire is made of heat and corrosion resistant steel wire 0.025 inch diameter.

(2) Lockwire must be tight after installation to prevent failure due to rubbing or vibration.

(3) Lockwire must be installed in a manner that tends to tighten and keep a part locked in place thus counteracting the natural tendency of the part to loosen.

(4) Lockwire must never be overstressed. It will break under vibration if twisted too tightly. The lockwire shall be pulled taut when being twisted, but shall have minimum tension, if any when secured.

(5) Lockwire ends must be bent toward the engine, or part, to avoid sharp or projecting ends which might present a safety hazard or vibrate in the air stream.

(6) Internal wiring must not cross over or obstruct a flow passage when an alternate method can be used.

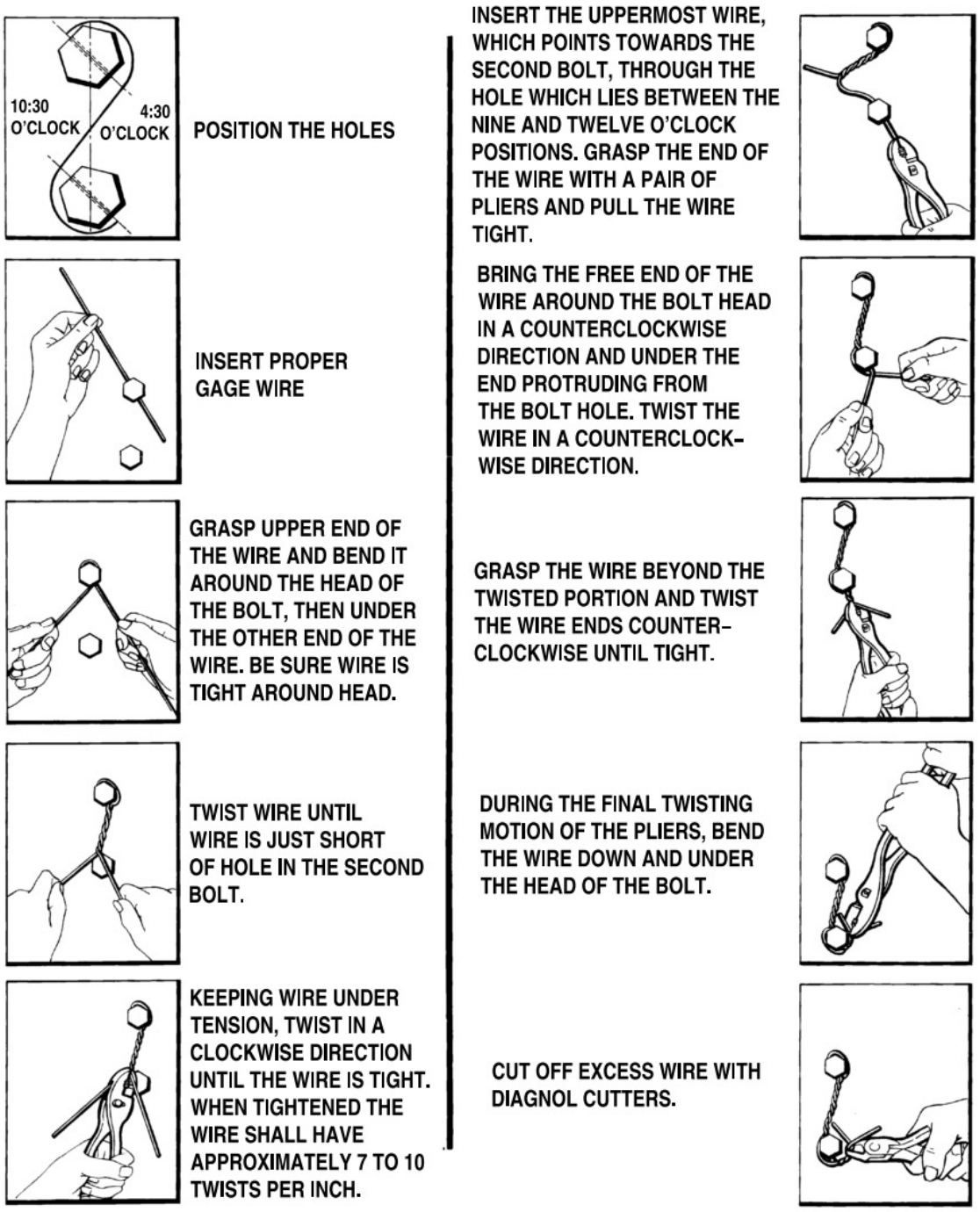
(7) Check the units to be lockwired to make sure they have been correctly torqued and that the wiring holes are properly positioned in relation to each other. When there are two or more units, it is desirable that the holes in the units be in the same relationship to each other. Never overtorque or loosen units to obtain proper alignment of the holes. It should be possible to align the wiring holes when the units are torqued within the specified limits. However, if it is impossible to obtain a proper alignment of the holes without either over or under torquing, select another unit which will permit proper alignment within the specified torque limits.

(8) To prevent mutilations of the twisted section of the wire when using pliers, grasp the wire at the ends or at a point that will not be twisted. Lockwire must not be nicked, kinked, or mutilated. Never twist the wire ends off with the pliers and when cutting off ends, leave at least three complete turns after the loop, exercising extreme care to prevent the wire ends from falling into the engine. The strength of the lockwire holes is marginal; never twist the wire off with pliers. Cut the lockwire close to the hole, exercising extreme care.

(9) Figure 204 illustrates a typical lockwiring procedure. Although there are numerous lockwiring operations performed on these engines, practically all are derived from the basic examples shown in Figure 205.

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Figure 204 Lockwiring Procedure

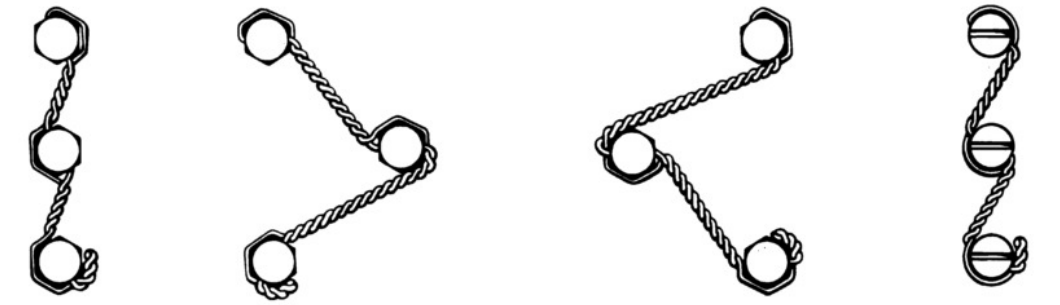


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Figure 205 Lockwiring Examples

(SHEET 1 OF 3)



EXAMPLE 1 EXAMPLE 2 EXAMPLE 3 EXAMPLE 4

EXAMPLES 1,2,3 AND 4 APPLY TO ALL TYPES OF BOLTS, FILLISTER HEAD SCREWS, SQUARE HEAD PLUGS, AND OTHER SIMILAR PARTS WHICH ARE WIRED SO THAT THE LOOSENING TENDENCY OF EITHER PART IS COUNTERACTED BY TIGHTENING OF THE OTHER PART. THE DIRECTION OF TWIST, FROM THE SECOND TO THE THIRD UNIT, IS COUNTER–CLOCKWISE TO KEEP THE LOOP IN POSITION AGAINST THE HEAD OF THE BOLT. THE WIRE ENTERING THE HOLE IN THE THIRD UNIT WILL BE THE LOWER WIRE AND BY MAKING A COUNTER–CLOCKWISE TWIST AFTER IT LEAVES THE HOLE, THE LOOP WILL BE SECURED IN PLACE AROUND THE HEAD OF THAT BOLT.



EXAMPLE 5 EXAMPLE 6 EXAMPLE 7 EXAMPLE 8

EXAMPLES 5,6,7 AND 8 SHOW METHODS FOR WIRING VARIOUS STANDARD ITEMS. WIRE MAY BE WRAPPED OVER THE UNIT RATHER THAN AROUND IT WHEN WIRING CASTELLATED NUTS OR ON OTHER ITEMS WHEN THERE IS A CLEARANCE PROBLEM.



EXAMPLE 9

EXAMPLE 9 SHOWS THE METHOD FOR WIRING BOLTS IN DIFFERENT PLANES. NOTE THAT WIRE SHOULD ALWAYS BE APPLIED SO THAT TENSION IS IN THE TIGHTENING DIRECTION.

EXAMPLE 10

EXAMPLE 10 SHOWS HOLLOW HEAD PLUGS WIRED WITH THE TAB BENT INSIDE THE HOLE TO AVOID SNAGS AND POSSIBLE INJURY TO PERSONNEL WORKING ON THE ENGINE.

EXAMPLE 11

EXAMPLE 11 SHOWS CORRECT APPLICATION OF SINGLE WIRE TO CLOSELY SPACED MULTIPLE GROUP.

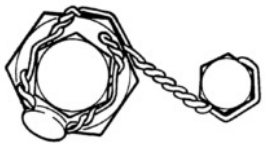
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(SHEET 2 OF 3)



EXAMPLE 12



EXAMPLE 13

EXAMPLES 12 AND 13 SHOW METHODS FOR ATTACHING LEAD SEAL TO PROTECT CRITICAL ADJUSTMENTS.



EXAMPLE 14

EXAMPLE 14 SHOWS BOLT WIRED TO A RIGHT ANGLE BRACKET WITH THE WIRE WRAPPED AROUND THE BRACKET.



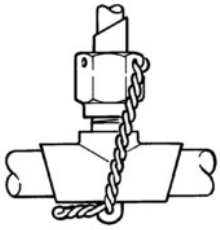
EXAMPLE 15

EXAMPLE 15 SHOWS CORRECT METHOD FOR WIRING ADJUSTABLE CONNECTING ROD.



EXAMPLE 16

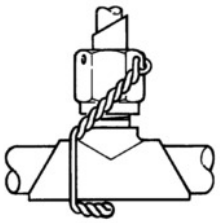
EXAMPLE 16 SHOWS CORRECT METHOD FOR WIRING THE COUPLING NUT ON FLEXIBLE LINE TO THE STRAIGHT CONNECTOR BRAZED ON RIGID TUBE.



EXAMPLE 17



EXAMPLE 18



EXAMPLE 19



EXAMPLE 20

FITTINGS INCORPORATING WIRE LUGS SHALL BE WIRED AS SHOWN IN EXAMPLES 17 AND 18. WHERE NO LOCKWIRE LUG IS PROVIDED, WIRE SHOULD BE APPLIED AS SHOWN IN EXAMPLES 19 AND 20 WITH CAUTION BEING EXERTED TO ENSURE THAT WIRE IS WRAPPED TIGHTLY AROUND THE FITTING.



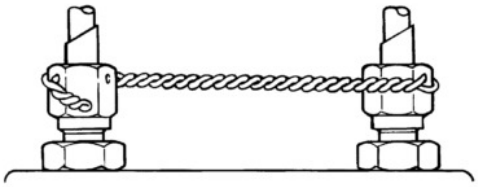
EXAMPLE 21

SMALL SIZE COUPLING NUTS SHALL BE WIRED BY WRAPPING THE WIRE AROUND THE NUT AND INSERTING IT THROUGH THE HOLES AS SHOWN.

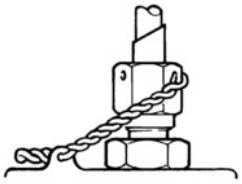
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(SHEET 3 OF 3)

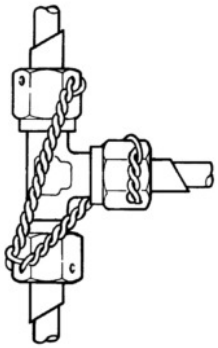


EXAMPLE 22



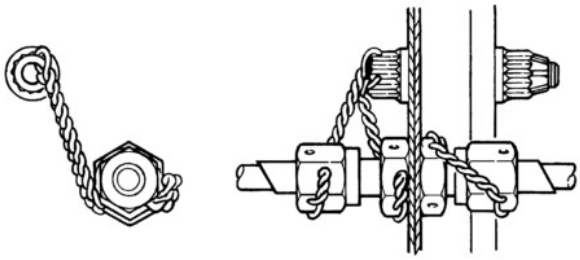
EXAMPLE 23

COUPLING NUTS ATTACHED TO STRAIGHT CONNECTORS SHALL BE WIRED AS SHOWN WHEN HEX IS AN INTEGRAL PART OF THE CONNECTOR.



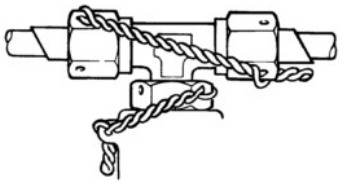
EXAMPLE 24

COUPLING NUTS ON A TEE SHALL BE WIRED AS SHOWN ABOVE SO THAT TENSION IS ALWAYS IN THE TIGHTENING DIRECTION.

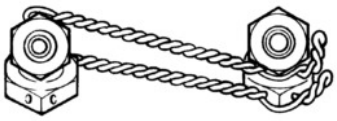


EXAMPLE 25

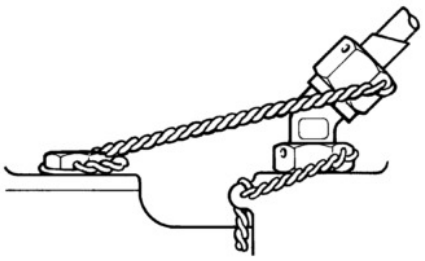
STRAIGHT CONNECTOR (BULKHEAD TYPE)



EXAMPLE 26



EXAMPLE 27



EXAMPLE 28

EXAMPLES 26, 27 AND 28 SHOW THE VARIOUS STANDARD FITTINGS WITH CHECK NUT WIRED SO THAT IT NEED NOT BE DISTURBED WHEN REMOVING THE COUPLING NUT.

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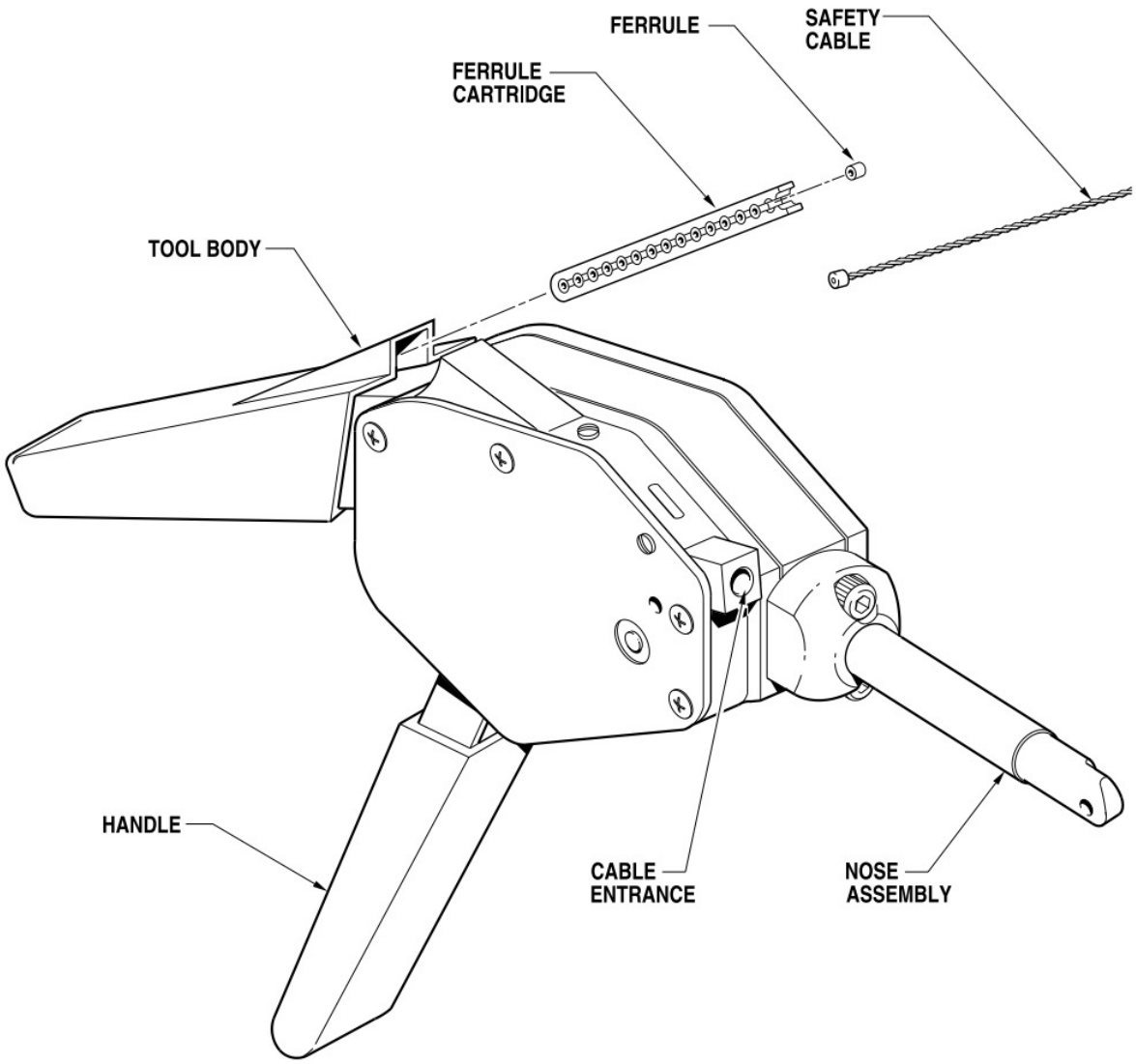
D. Safety Cable (Ref. Figs. 206 and 207)

(1) General:

(a) Safety cable is installed through two or more parts in such a way that as the fastener loosens the safety cable will tighten. By tightening, the safety cable will not permit the fastener to loosen or turn. The safety cable system is made up of two components: the safety cable/ferrule kit (PWC05-344) and a crimping tool (PWC90025) (Ref. Fig. 206).

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Figure 206 Standard Safety Cable Components



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(b) The safety cable is available in one size, 0.032 inch (0.81 mm) diameter, and is made of stainless steel material. One end of the cable will have a cable end fitting swaged to it. The cable end fitting is made of stainless steel material. The strands of the cable on the opposite end of the cable are fused together to prevent the cable from fraying.

(c) The ferrules are made of stainless steel material and are purchased in a spring-loaded, disposable cartridge. When the safety cable is installed the ferrule will be crimped on the open end of the cable.

(d) The hand operated crimping tool comes in two different lengths. The crimping pressure of the tool is set by the manufacturer. The primary parts of the crimping tool are the nose assembly and the tool body. This tool is used to crimp the ferrule on the end of the safety cable. The crimping tool will cut the safety cable even with the ferrule at the same time the ferrule is crimped.

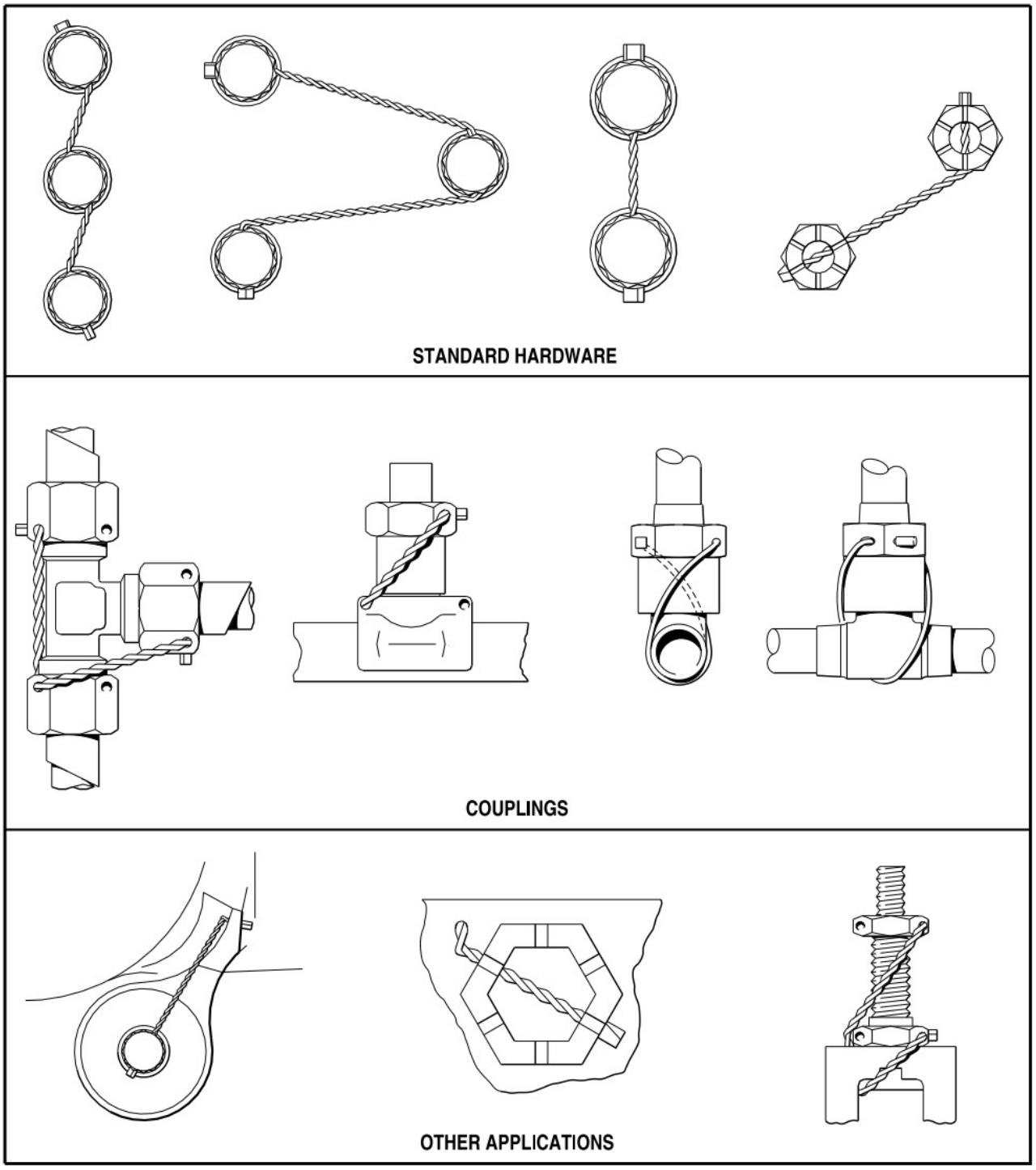
(e) Where safety cable is specified, it may be replaced with lockwire. Use lockwire as specified in Table 203.

Table 203 Safety Cable Substitution

Nominal Safety Cable Diameter	Lockwire Substitution
0.032 inch (0.815 mm)	AS3214-02 or MS9226-04
NOTE: Lockwire AS3214 may be used in engine external applications only.	

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Figure 207 Safety Cable Patterns



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(2) Basic Installation Rules:

(a) It is not permissible to re-use safety cable or a ferrule. The safety cable and ferrule must be new for every application.

(b) Unless specified differently in assembly procedure, safety cable should be installed in double- or triple-bolt patterns. The double-bolt pattern is preferred when you apply safety cable to an even number of fasteners.

(c) The maximum allowable span of safety cable between two end points is 6 inches (152.4 mm), unless otherwise specified.

(d) Any damages (such as fray, nicks, kink or any damage to the safety cable) to any portion of the safety cable prior to, during or after installation is not permitted.

CAUTION: DO NOT USE SAFETY CABLE IF THE HOLES ARE MORE THAN 0.100 INCH (2.54 mm) DIAMETER. THE FERRULES ON THE END OF THE CABLE ARE 0.106 - 0.108 INCH (2.69 - 2.74 mm) DIAMETER AND WILL FALL THROUGH HOLES LARGER THAN THIS.

(e) Always install safety cable through the safety cable holes provided on the fastener.

(f) Do not decrease or increase the torque on the parts to correctly align the holes for the cable.

(g) Install safety cable so that any tendency for a fastener to loosen is counteracted by more tension on the cable.

(h) It is recommended that there be no sharp turns more than 90° when the cable is passed through a fastener. Installation of safety cable should be with either a positive or neutral pull.

(i) Cable flex limits are specified in Figure 210. These values are the maximum flex limits between the end points when light finger pressure of approximately 2 pound (8.9 N) is applied at midspan.

(j) Cut off excess cable from crimped ferrule after installation of safety cable.

(k) Maximum length of cable beyond crimped ferrule is 0.031 inch (0.787 mm).

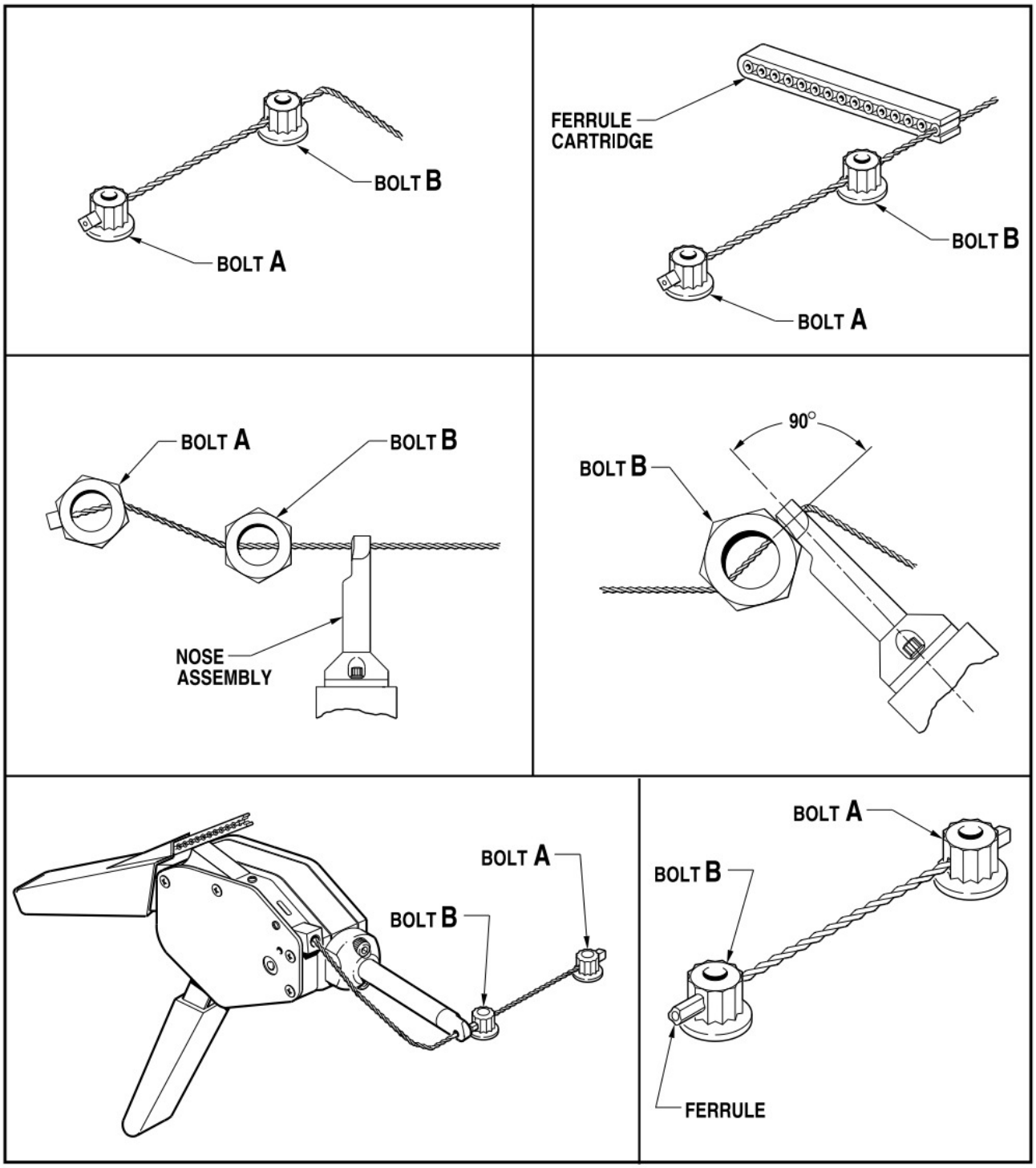
(l) Where specified, apply and crimp an aluminum seal to the load bearing portion of the safety cable.

(m) After installing the seal, make sure that it is snug on the cable and that it cannot be moved without showing evidence of tampering.

(n) After installing the seal, an inspection stamp or other means of identification shall be applied to the seal.

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Figure 208 Safety Cable Installation



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(3) Installation (Ref. Fig. 208):

(a) Install ferrule cartridge into crimper tool body (PWC90025).

(b) Do a visual inspection of the holes to be used to find all damage. If the hole is damaged, replace the part or, if possible, use another hole to install safety cable.

NOTE: Keep the safety cable as straight as possible when you locking the fasteners or parts together.

(c) Insert the free end of the safety cable (PWC05-344) into the hole of the part. Pull the cable through the hole until the cable end fitting is against the part.

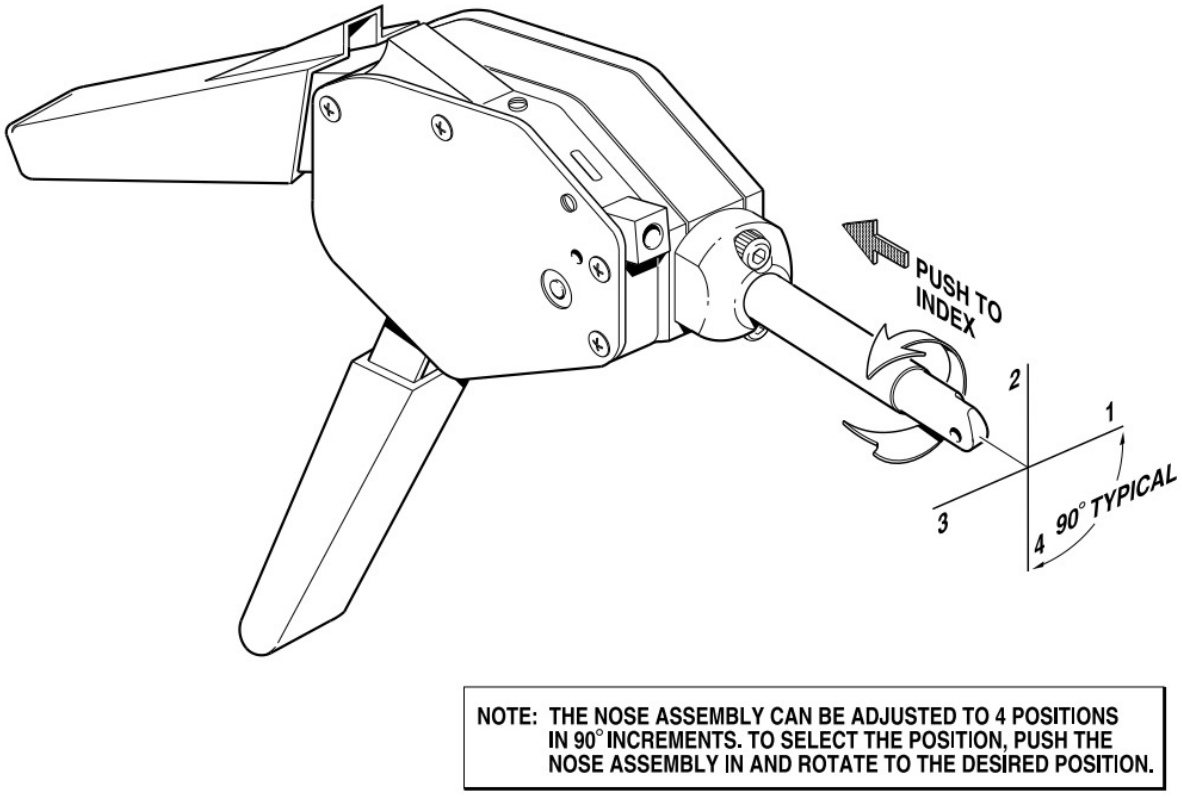
NOTE: On a double-bolt pattern, do not safety cable in a negative pull direction. Make sure the safety cable has a positive or neutral pull.

(d) Install the end of the cable through the second part. Choose the hole in the part that permits the cable to be as straight as possible. Pull the cable through the second part. If three parts must be safety cabled, do the same procedure for the third part.

(e) The nose can be changed to four different positions at 90° increments. To select the position, push the nose in towards the tool, rotate to the desired position, and release the nose. If the nose does not lock into the indexed position, turn it slightly until it does (Ref. Fig. 209).

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Figure 209 Nose Assembly Rotation



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(f) Install the free end of the safety cable through a ferrule (PWC05-344). Use the safety cable to pull the ferrule out of the ferrule cartridge.

(g) Put the end of the safety cable through the crimping head of the crimping tool. Make sure the large hole in the crimping head is on the same side as the ferrule. Move the crimping tool along the safety cable until the crimping head is against the part. Make sure the ferrule goes inside the hole in the crimping head.

(h) Install the free end of the cable into the cable entrance. Continue to push the cable into the cavity. When the free end of the cable appears at the bottom of the tool, grab the cable and pull the slack from the cable until resistance is felt.

NOTE: Grip the cable as close to the free end as possible during insertion to prevent buckling of the cable.

(i) When all slack is removed from the cable, make sure the tool is snugly against the fastener by using several short strokes of the handle. Release the handle to the full open position and fully close the handle to affect crimping and cutting.

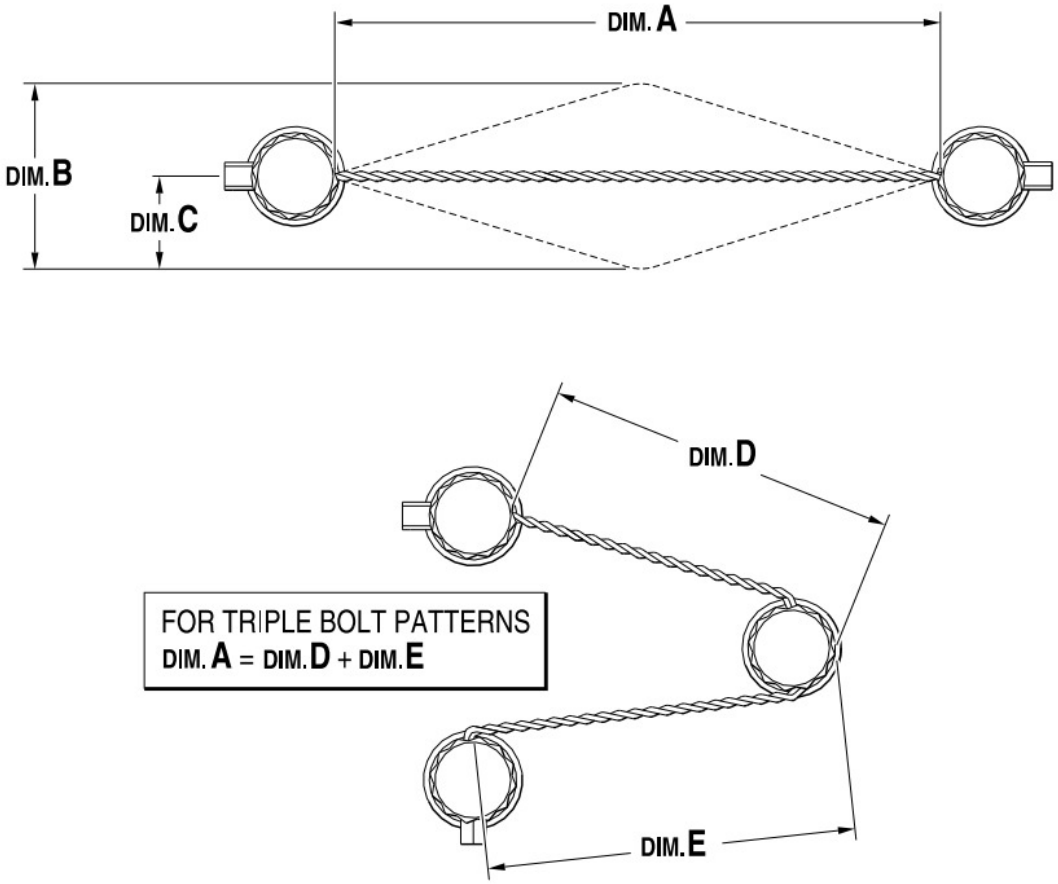
CAUTION: IT IS IMPORTANT ON THIS FINAL STROKE TO HOLD THE TOOL AS STEADY AND PERPENDICULAR TO THE CABLE AS POSSIBLE WHILE ENSURING THAT A FULL STROKE IS USED TO MAINTAIN A CONSISTENT TENSIONING OF THE CABLE.

(j) After crimping and cutting, release the tool handle and slide the tool off the crimped ferrule. Pull the excess cable from the tool.

(k) Remove and discard unused safety cable from the bottom of the crimping tool.

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Figure 210 Safety Cable Flex Limits



DIM. A	DIM. B	DIM. C
0.5 IN. (12.7 mm)	0.125 IN. (3.18 mm)	0.062 IN. (1.59 mm)
1.0 IN. (25.4 mm)	0.250 IN. (6.35 mm)	0.125 IN. (3.18 mm)
2.0 IN. (50.8 mm)	0.375 IN. (9.52 mm)	0.188 IN. (4.76 mm)
3.0 IN. (76.2 mm)	0.375 IN. (9.52 mm)	0.188 IN. (4.76 mm)
4.0 IN. (101.6 mm)	0.500 IN. (12.70 mm)	0.250 IN. (6.35 mm)
5.0 IN. (127.0 mm)	0.500 IN. (12.70 mm)	0.250 IN. (6.35 mm)
6.0 IN. (152.4 mm)	0.625 IN. (15.88 mm)	0.312 IN. (7.94 mm)

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(4) Verification (Ref. Fig. 210):

(a) Push against the safety cable with light finger pressure halfway between the locked parts. If the cable feels loose, do a dimensional check to make sure the safety cable is serviceable as follows:

1 Measure the distance between the locked parts. Write this measurement down as Dimension A. If three parts are locked together, measure the distance between each of the parts and add the two measurements together to get Dimension A.

2 Push against the safety cable with light finger pressure halfway between the two locked parts. Measure the distance the safety cable moves laterally. Writes this measurement down as Dimension C.

3 Compare the dimensions written down to the limits given.

CAUTION: DO NOT TRY TO BREAK THE SAFETY CABLE. IF THE SAFETY CABLE MUST BE REMOVED, CUT THE SAFETY CABLE TO AVOID DAMAGE TO THE HOLES OF THE PART.

(b) If the safety cable is not within limits, cut the safety cable with wire cutters and remove the safety cable. Install new safety cable.

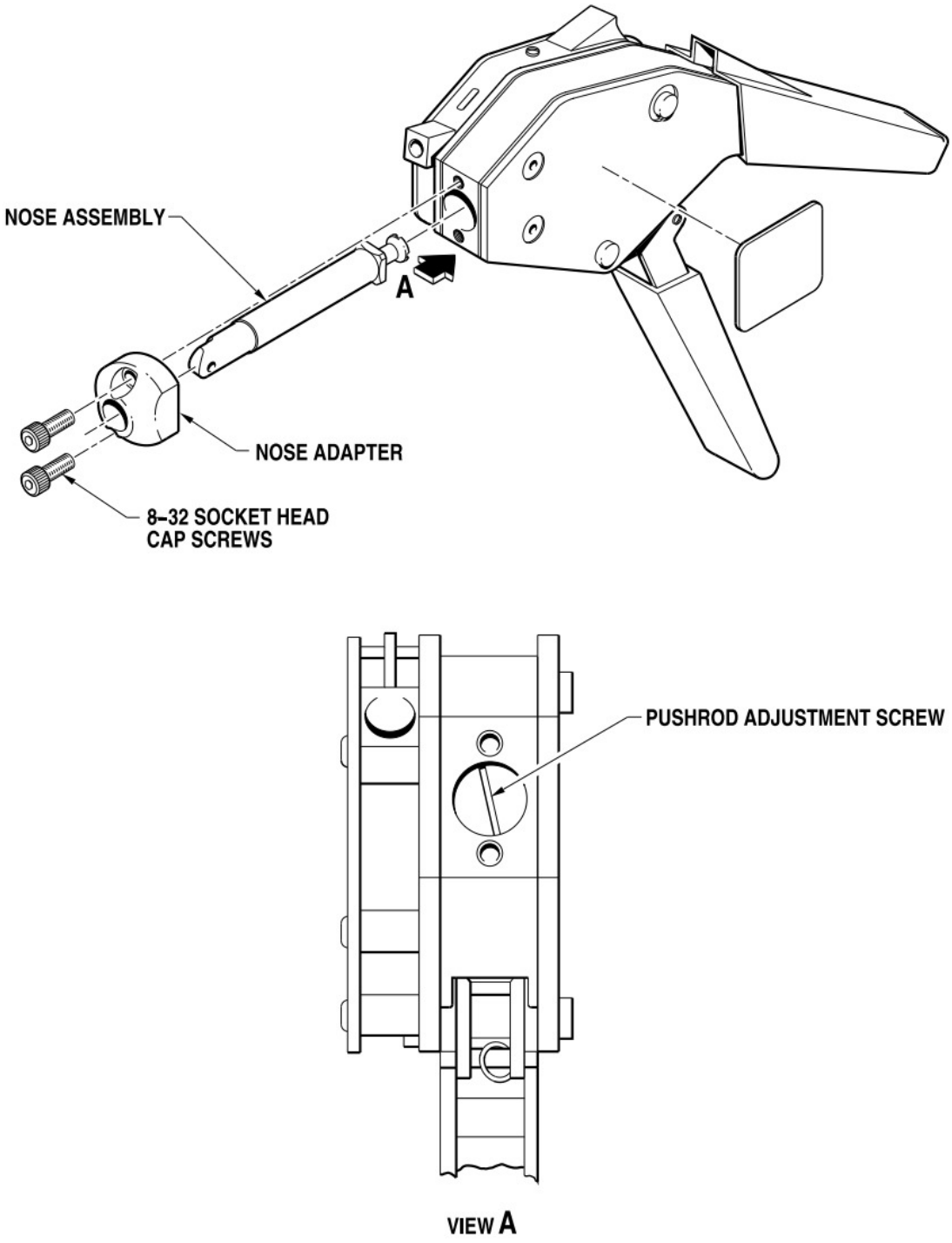
(c) Check that cable was installed through bolts correctly (Ref. Step (3)).

(d) Verify that cable was cut flush at the end of the ferrule with no strand extending more than 1/32 inch from the end of the ferrule.

(e) Visually inspect safety cable for kinks, frayed wires, or improper crimps. Remove and replace safety cable if problem exists.

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Figure 211 Indenter Adjustment



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(5) Indenter Adjustment (Ref. Fig. 211):

(a) Remove the nose assembly by removing the two socket head cap screws.

CAUTION: DO NOT ADJUST THE PUSHROD MORE THAN A QUARTER OF A TURN AT A TIME. SEVERE ADJUSTMENTS MAY DAMAGE THE TOOL.

(b) Adjust the pushrod adjustment screw using a 0.25 inch (6.4 mm) straight edge screw driver. Rotate the pushrod adjustment screw clockwise to loosen the crimp (enlarge the gaging dimension). Rotate the adjustment screw counterclockwise to tighten the crimp (reduce gaging dimension).

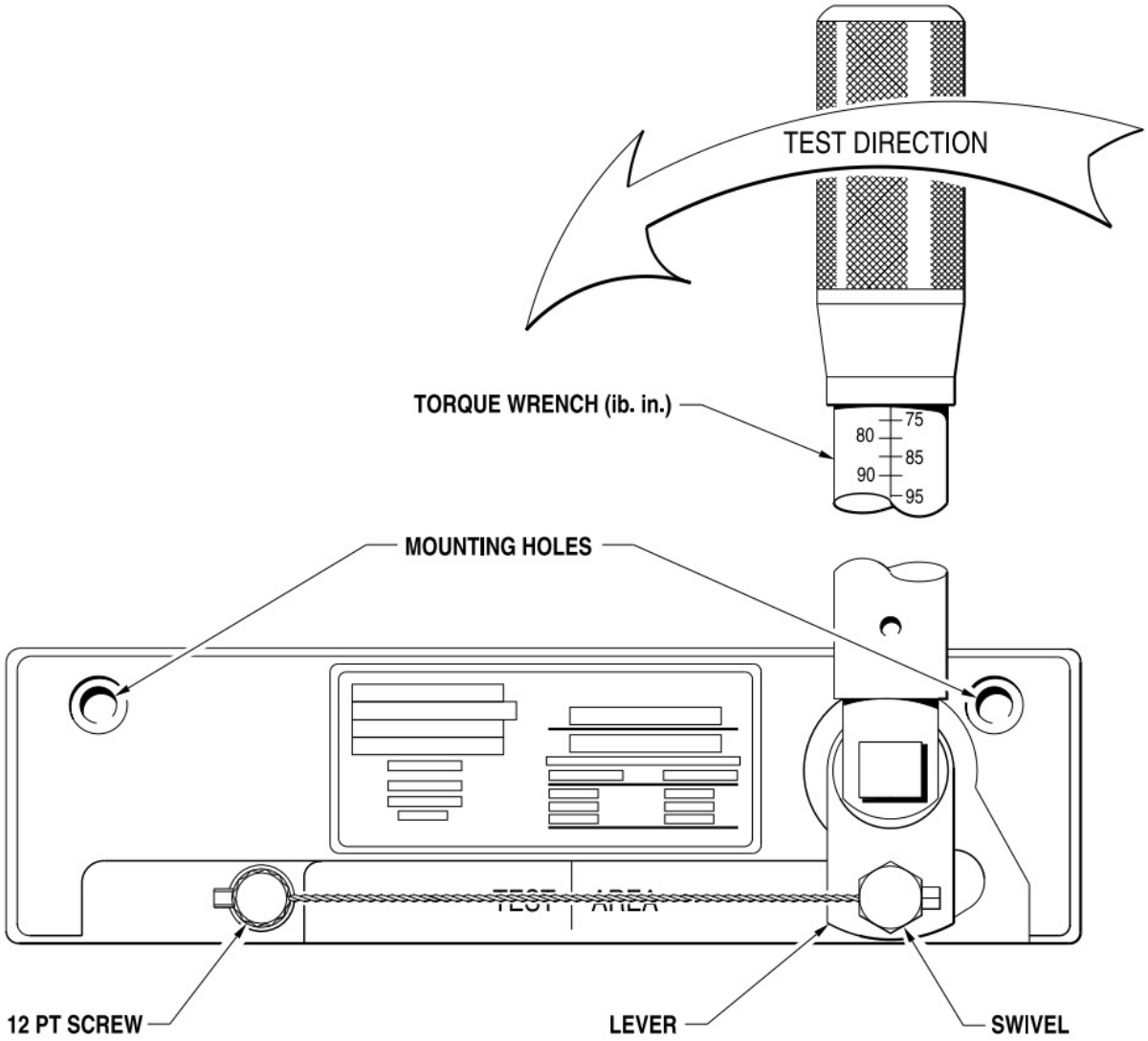
(c) Replace the nose assembly and the two socket head cap screws. Rotate each screw a small amount until tight to prevent binding.

CAUTION: TIGHTENING THE CRIMP INCREASES REQUIRED HAND FORCES DURING APPLICATION AND INCREASES CABLE ASSEMBLY TENSILE VALUES. LOOSENING THE CRIMP DECREASES REQUIRED HAND FORCES DURING APPLICATION AND DECREASES CABLE ASSEMBLY TENSILE VALUES.

(d) Check calibration as described in (Ref. Step. (6)).

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Figure 212 Test Block - Crimper Calibration



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(6) Indenter Calibration (Ref. Fig. 212):

NOTE: 1. Indenter calibration should be checked periodically and must be checked after the nose assembly has been removed or interchanged.

NOTE: 2. Do a pull-off load test as often as necessary to make sure the crimp done by the crimping tool meets the necessary requirements.

(a) Set up test block (PWC90026).

(b) Thread the cable into the 12 point screw and through the swivel, holding the lever in the home position. If the lever is not kept in the home position the results may be adversely affected. Cut the cable by making sure to close the handles completely.

(c) Apply approximately 2 lbs. force to the cable with your finger at the line marked test area. If the cable touches either the side or bottom surface of the test area then remove the cable from the test block and use another cable as described (Ref. Step. (6)(b)).

(d) Place a 3/8 inch drive torque wrench into the square drive hole, orientating the test block and torque wrench on a flat surface. Apply the proper pull-off load as shown in Table 204. Release torque when minimum pull-off load is achieved.

NOTE: Do not apply additional tension to safety cable than that is required to achieve minimum pull-off load.

(e) If the cable did not break or pull out of the ferrule after applying proper torque, remove the torque wrench and apply approximately 2 lbs. force to the cable with your finger at the line marked test area.

(f) If the cable breaks, touches either the side or bottom surface of the test area or pulls out of the ferrule, then the crimp is unacceptable and the tool indenter should be adjusted (Ref. Step. (5)).

Table 204 Safety Cable Minimum Crimp Requirements (Pull-off Load)

Nominal Cable Diameter	Minimum Pull-off Load
0.032 Inch (0.81mm)	70 Inch-Pounds (7.909 Nm)

(7) Maintenance of Crimp Cavity:

(a) Debris can accumulate in the crimp cavity during use. This debris must be removed and the cavity oiled periodically depending upon use and environment.

NOTE: Ferrule sticking can occur if this procedure is not followed.

(b) Remove debris by gently scraping or brushing the crimp cavity.

(c) Blow any remaining debris from the cavity with air.

(d) Coat the cavity with a light film of any lightweight synthetic or petroleum based oil after the cavity has been thoroughly cleaned.

7. Identification of Hardware Material, Particularly Nuts and Bolts

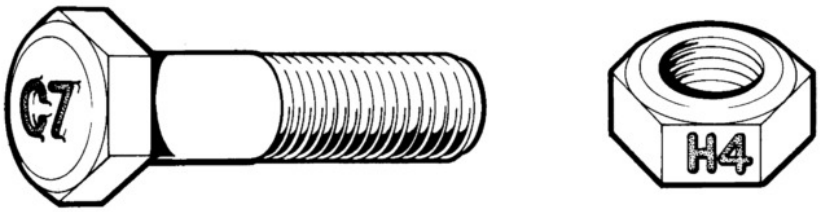
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A. A permanent type of material code designation has been adopted for use in these engines. Correct engine reassembly procedures require that particular attention be paid to the material requirements for nuts and bolts used in the hot section of the engine. In these areas, where parts must be of material which is resistant to high operating temperatures, special heat resistant alloys are employed. It is imperative, that at reassembly of the engine or its components, the properly coded part be reassembled in its original location, if serviceable.

B. The code system employs the use of a letter "C" for corrosion resistant steel for normal application and "H" for heat resistant alloys in hot section application. The stamped or embossed letter will be followed by a number of one or more digits, such as C1, C8, H3, and H12. Bolt code identification will usually appear on the top of the head and nut identification on one side of the hex (Ref. Fig. 213). When the application is an AN or MS six digit part number, the code identification "C" or "H" will be preceded by the letter "E" as in EC3 or EH10.

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Figure 213 Hardware Code Identification



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NOTE: All AN or MS six digit part numbers, when manufactured of material in the common temperature range (such as cadmium plated, low alloy steel parts), are also coded "E" to indicate, in part, close material quality control.

CAUTION: NEVER ASSEMBLE A MATERIAL CODED PART WITH A "C" IDENTIFICATION IN A LOCATION WHICH REQUIRES AN "H" CODED PART AND VICE-VERSA. NEVER USE CADMIUM PLATED NUTS, BOLTS OR ANY OTHER CADMIUM PLATED PART IN THE HOT SECTION OF THE ENGINE.

C. The adoption of this program will make it possible for service activities to avoid the assembly of parts, with similar physical appearance, in locations which require high heat resistant parts. In this regard, it is required, at time of disassembly (removal), that all similarly coded parts be segregated so that two, or more, physically similar parts with different code numbers will not be scrambled; and so that, at subsequent reassembly (installation), the properly coded parts are replaced in their proper locations.

D. Parts from several engines should not be scrambled at disassembly (removal) and cleaning, because, at the time of engine manufacture, production requirements may necessitate one or more material substitutions in a block of engines. Subsequent engine reassembly with indiscriminate scrambling of hardware, may result in unnecessary confusion in sorting these parts for proper reassembly.

CAUTION: IT IS IMPORTANT THAT "J" THREADED HARDWARE BE CAREFULLY SEGREGATED AND MARKED AT DISASSEMBLY (REMOVAL) TO PREVENT MIXING WITH NON "J" THREADED PARTS.

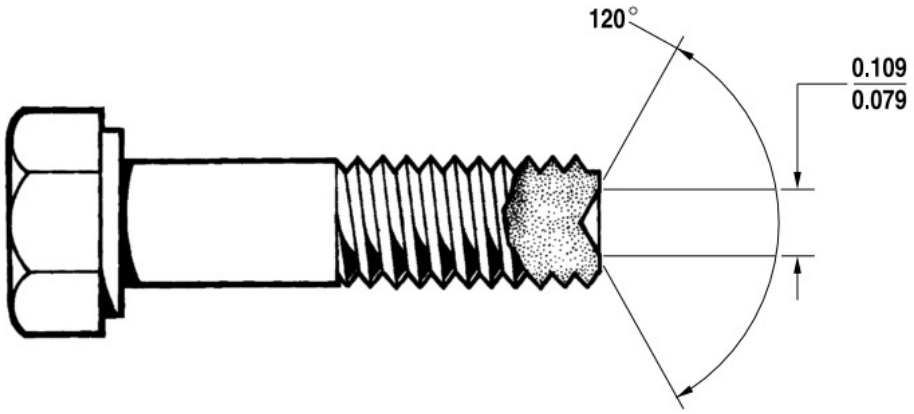
E. Unified Controlled Radius Root "J" Threaded Bolts feature full root radius for external threads with increased minor internal threaded diameter. "J" threads ensure greater fatigue life, improved stress relaxation characteristics and show improvement in tensile strength over previous designs. This thread form is used extensively throughout engines. Engine bolts having "J" threads appear just like items threaded with conventional threads.

8. Stainless Steel Bolts with Reduced Pitch Diameters

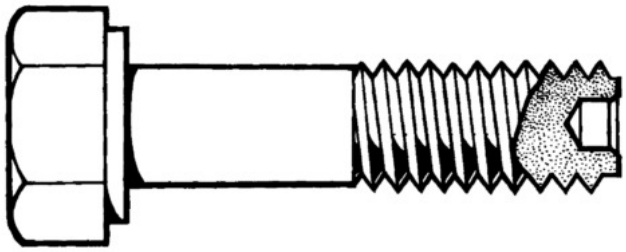
At engine manufacture, it is the practice to provide bolts with reduced pitch diameter for use in the hot sections of engines. This will minimize the possibility of bolt and nut seizure and is standard practice for locations where the parts are subjected to elevated temperatures (Ref. Fig. 214).

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Figure 214 Reduced Pitch Diameter Bolt Identification



REDUCED PITCH DIAMETER BOLT



STANDARD BOLT

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9. Debris Analysis and Material Specifications

A. General

(1) Debris analysis monitors wear of oil-wetted engine parts. This procedure improves aircraft serviceability and dispatch reliability and reduces engine repair costs. This is achieved by identifying potential engine problems at the earliest possible stage, thus minimizing the possibility of in-flight shutdowns, away-from-base engine changes and secondary damage.

(2) Rapid wear or surface fatigue (e.g. on gear teeth contact surfaces, bearing raceways, rolling elements, housing bearing bores, etc.) or interference between rotating and fixed components (e.g. oil pump pinions, associated housings and labyrinth seals, etc.) produces debris. Magnetic debris over 100 microns in diameter is usually captured by the chip detectors. Magnetic and non-magnetic debris over 10 microns in diameter is usually captured by the oil system filters. For engine equipped with such devices, it is also possible to find debris on the magnetic pulse pick-up sensors.

(3) Analysis (form, appearance, dimensions, quantity and material) of the debris generated by oil wetted parts is necessary to provide the information needed to facilitate locating the source of the debris and determining any preventive maintenance action required.

B. Filter Patch Check Debris Inspection/Analysis

(1) The analysis should be done by a qualified laboratory (Ref. Para . F.). Operators may send the contaminated filter or the debris to the laboratory for the filter patch check and/or analysis and provide the following information:

(a) Engine model and serial number.

(b) Engine time since new (TSN) or time since overhaul or refurbishment (TSO or TSR).

(c) Time since last filter inspection.

(d) Filter location.

(e) Reason for filter removal, inspection and debris analysis.

(2) Reports supplied by the laboratory to the operator containing the results of the analysis must include:

(a) Information provided by the operator with the filter or debris.

(b) Total weight of debris.

(c) Material and amount of non-magnetic debris when classified as major (Ref. NOTE).

(d) Material, appearance, shape and amount of non-magnetic metallic debris.

(e) Material, appearance, shape and amount of metallic debris.

NOTE: The amount of individual constituents in the debris should be classified as:

- Major - when the weight of the constituent is more than 50% of the total debris weight.
- Minor - when the weight of the constituent is less than 50% of the total

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debris weight.

- Traces - when the weight of the constituent is less than 5% of the total debris weight.

(3) If bearing material (Ref. Table 205) is found, the operator must be advised as soon as possible by telephone and provided with a detailed written report.

C. Inspection/Analysis of Debris Collected on Chip Detector or Magnetic Pulse Pick-up Sensors

(1) The analysis should be done by a qualified laboratory (Ref. Para . F.).

(2) The operator must provide the laboratory with the following information:

(a) Engine model and serial number.

(b) Engine time since new (TSN) or time since overhaul or refurbishment (TSO or TSR).

(c) Locations where debris were found.

(d) Reason for inspection (scheduled or unscheduled).

(3) Reports supplied by the laboratory to the operator containing the results of the analysis must include:

(a) Information provided by the operator when submitting the material for analysis.

(b) Type(s) of material found.

(c) Shape and appearance of the material.

(d) If bearing material (Ref. Table 205) is found, the operator must be advised as soon as possible by telephone and provided with a detailed written report.

D. Identification of Metal Particles

(1) General

(a) When unidentified particles of metal are found, they may be either steel, tin, aluminum, magnesium, silver, bronze, titanium, or cadmium. In some cases the type of metal may be determined by the color and hardness of the pieces. However, when the particles cannot be positively identified by visual inspection and knowledge of the exact character of the metal is desired as an aid to troubleshooting, a few simple tests will determine the kind of metal present.

WARNING: USE EXTREME CARE IN HANDLING ACIDS.

(b) The following equipment and chemicals are required to make these tests:

a source of open flame

permanent magnet

two ounces of solution containing ten percent ammonium nitrate(PWC05-011) and water

electrical soldering iron

two ounces each of 50 percent by volume hydrochloric acid (PWC05-196) or (PWC05-

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196A) and concentrated nitric acid (PWC05-195)

sodium hydroxide pellets (PWC05-056)

watch glass

white porcelain spot plate

ammonium biflouride crystals (PWC05-009)

5 to 10 percent hydrofluoric acid (PWC05-004) or concentrated sulfuric acid (PWC05-074)

3 to 10 percent hydrogen peroxide (PWC05-038)

concentrated phosphoric acid (PWC05-006)

(2) Procedure

NOTE: The following test procedure is recommended for determining the character of unknown metal particles. For best results, follow the steps as outlined:

(a) Steel - the particles of many steels can be isolated by means of the permanent magnet. Magnetic steel or iron is attracted by the magnet.

(b) Titanium:

WARNING: THE FOLLOWING CHEMICALS ARE HAZARDOUS AND REQUIRE SPECIAL HANDLING. IT IS RECOMMENDED THE FOLLOWING TESTS BE DONE BY QUALIFIED LABORATORY PERSONNEL.

1 Place a piece, or pieces of the metal to be identified, on a white porcelain spot plate. A piece of titanium or titanium-bearing metal should be placed on another spot plate to observe and verify the results obtained.

2 Add several crystals of ammonium bifluoride (PWC05-009) and 5 to 10 drops of water to the metal particles. (Two or three drops of a 5 to 10 percent hydrofluoric acid solution (PWC05-004) can be used instead).

NOTE: Solid ammonium bifluoride is crystalline and can be conveniently stored in a dry area and used as required.

3 Let stand 20 to 30 minutes, or until the solution becomes slightly discolored.

4 Add two or three drops of 1:1 sulfuric acid (one part demineralized water (PWC05-073) to one part concentrated acid PWC05-074).

5 Let stand 20 to 30 minutes, or until solution becomes more discolored.

6 Add three to four drops of 3 to 10 percent hydrogen peroxide (PWC05-038). Solution must not be too old.

7 If titanium is present, a yellowish color will develop. This yellow color will become progressively darker with time, if allowed to set.

8 Add two to three drops of concentrated phosphoric acid (PWC05-006), and stir to discharge any yellow color due to the possible presence of iron.

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9 Any light yellow to orange coloration indicates the presence of titanium.

WARNING: NEVER TRY TO BURN MORE THAN A FEW PARTICLES OF METAL SUSPECTED TO BE MAGNESIUM. DUST IS EXPLOSIVE.

(c) Magnesium - a simple test for these particles is burning. Magnesium will burn with a bright white flash.

(d) Cadmium - place the remaining particles in the solution of water and ammonium nitrate(PWC05-011). If all or any of the particles dissolve in this solution, they are cadmium. After this test, rinse and dry any remaining particles.

(e) Tin - the tin particles can be distinguished by their low melting point. With a clean soldering iron, heated to 260°C (500°F) and tinned with 50-50 solder (50 percent tin - 50 percent lead), a tin particle dropped on the iron will melt and fuse with the solder.

(f) Aluminum - when a particle of aluminum is placed in hydrochloric acid (PWC05-196), 50 percent by volume, it will fizz with rapid emission of gas bubbles and gradually disintegrate and form a black residue (aluminum chloride). Silver and bronze do not noticeably react with hydrochloric acid.

(g) Aluminum Paint - use this procedure to determine whether or not the material is aluminum silicone paint, aluminum chips, or silver particles.

1 Make a solution by adding one pellet of sodium hydroxide (PWC05-056) to three ccs of water.

2 Place several drops of this solution on a watch glass and drop in the suspected particles.

3 If the particles are aluminum silicone paint, there will be a mild reaction in the form of gas bubbles and some visible gas as the particles change to sodium aluminate.

4 If the particles are aluminum chips, the reaction will be much more active with many more gas bubbles forming and more visible gas.

5 If the particles are silver, there will be no reaction.

(h) Silver - when a silver particle is placed in nitric acid, it reacts rather slowly, producing a whitish fog in the acid.

(i) Bronze - when a bronze (or copper) particle is placed in nitric acid, a bright green cloud is produced.

E. Material Specifications

(1) To facilitate the identification of components which are the source of debris found in the oil system, the material specifications are listed in Table 205 . The common contaminants found in the oil system are listed in Tables 206 and 207.

Table 205 Material Specification (Engine Components)

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Specification	Material	Nomenclature	Engine Location
AMS 2412	Silver Plating	Cage, No. 1 Brg	GG
		Cage, No. 2 Brg	GG
		Cage, No. 3 Brg	PT
		Cage, No. 4 Brig	PT
		Cage, No. 5 Brg	RGB
		Cage, No. 6 Brg	RGB
		Cage, No. 7 Brg	RGB
		Cage, FCU Drive Brg	AGB
		Cage, Start-Gen Drive Brg	AGB
AMS 4082, 4150	Alum (AA6061)	Sleeve, First-stage, Carrier	RGB
AMS 4117, 4127, 4150	Alum (AA6061)	Stator, No. 1 Brg Air Seal	GG
AMS 4214	Alum (Alloy 355)	Breather, Impeller	AGB
		Body, Main Scavenge Oil Pump	AGB
		Housing, Main Scavenge Oil Pump	AGB
		Cover, Main Scavenge Oil Pump	AGB
		Housing , External Scavenge Oil Pump	AGB
		Spacer, External Scavenge Oil Pump	AGB
		Cover Housing, Scavenge Oil Pump	AGB
AMS 4260	Alum. (Alloy 356)	Cover, Main Oil Pressure Pump	AGB
AMS 4275	Alum. (Alloy 750)	Brgs, Main Oil Pump Gear	AGB
		Sleeve, Accessory Drive Gears	RGB
AMS 4439	Magnesium (ZE41E)	Housing, Front	RGB

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Specification	Material	Nomenclature	Engine Location
		Housing, Rear	RGB
		Housing, Main Oil Pressure Pump	RGB
		Diaphragm	AGB
		Housing	AGB
AMS 4212	Alum (Alloy 355)	Compressor Inlet Case	GG
AMS 4616	Bronze	Cage, FCU Drive Brgs	AGB
		Cage, Start- Gen Drive Brg	AGB
		Cage, Vacuum Pump Drive Brg	AGB
		Cage, External Scavenge Oil Pump Drive Brg	AGB
		Coupling Shaft Brg	RGB
AMS 5315	Steel	Sleeve, Propeller, Oil Transfer	RGB
AMS 5613	SST (X51410)	No. 1 Brg Front Air Seal	GG
		No. 2 Brg Front Stator, Air Stator	GG
		No. 2 Brg Front Rotor Air Seal	GG
		No. 2 Brg Rear Stator Air Seal	GG
		No. 2 Brg Rear Rotor Air Seal	GG
		No. 3 Brg Stator Air Seal ,	PT
AMS 5645, 5646	SST (30321)	Sleeve, AGB Coupling Shaft Brg	AGB
AMS 5671	SST (Inconel X750)	Seal, No. 3 Brg, Front Labyrinth, Air	PT
AMS 5732	SST (A286)	Housing, No. 1 Brg, Outer	GG
AMS 6260	Steel (9310)	Gear, Main Oil Scavenge Pump	AGB
AMS 6265	Steel	Gear, Accessory Drive, Front	RGB

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Specification	Material (9310)	Nomenclature	Engine Location
		Gear, Second-stage Planet	RGB
		Carrier, Second-stage Planet Gear	RGB
		Shaft, Second-stage Planet Gear	RGB
		Gear, Ng Tacho Drive	RGB
		Gear, Propeller and O/S Governor Drive	RGB
		Coupling, Propeller Governor Drive	RGB
		Gear, Second-stage, Ring	RGB
		Gear, First-stage, Ring	RGB
		Gear, Second-stage, Sun	RGB
		Coupling, Second-stage Sun Gear	RGB
		Adapter, Second-stage Sun Gear	RGB
		Carrier, First-stage Planet Gear	RGB
		Gear, First-stage, Planet	RGB
		Gear, First-stage, Sun	RGB
		Coupling, AGB Drive	GG
		Coupling, First-stage Sun Gear	PT
		Shaft, AGB Coupling	AGB
		Gear, FCU, Drive	AGB
		Gear, Start- Gen Drive	AGB
		Gear, Vacuum Pump Drive	AGB
		Gear, Hydraulic Pump Drive	AGB
		Gear, Main Oil Pump Drive	AGB
		Gear, External Scavenge Pump Drive	AGB
		Gear, Oil Pressure Pump	AGB
		Gear, External Oil Scavenge Pump	AGB

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Specification	Material	Nomenclature	Engine Location
		Gear, Scavenge Oil Pump	AGB
		Shaft, Vacuum Pump Drive	AGB
		Shaft, External Scavenge Pump Drive,	AGB
		Coupling, Oil Pump Drive	AGB
AMS 6322	Steel (8740)	Sleeve, Second-stage Sun Gear	RGB
		Shaft, First-stage Planet Gear	RGB
AMS 6414	Steel (4340)	Cage, No. 7 Roller Brg	RGB
		Torque Piston	RGB
		Inner Housing No. 1 Brg	GG
		Cage, No. 1 Brg	GG
		Cage, No. 2 Brg	GG
		Cage, No. 3 Brg	PT
		Cage, No. 4 Brg	PT
		Coupling, Main Oil Pump Drive Gear	AGB
AMS 6415	Steel (4340)	Shaft, Propeller	RGB
		Cage, No. 6 Brg	RGB
		Cage, No. 5 Brg	RGB
		Locating Pin, Second-stage Planet Gear Carrier,	RGB
		Torque Cylinder,	RGB
		Shaft, Impeller	GG
AMS 6440/ 6444	Steel (52100)	Roller, AGB Coupling Shaft Brg	AGB
		Roller, Start-Gen Drive Brg	AGB
		Roller, Hydraulic Pump Drive Brg	AGB

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Specification	Material	Nomenclature	Engine Location
		Roller, Vacuum Pump Drive Brg	AGB
		Roller, External Scavenge Pump Drive Brg	AGB
		No. 7 Brg, Roller and Rings	RGB
		No. 6 Brg, Ball and Rings	RGB
		No. 5 Brg, Roller and Rings	RGB
		First-stage Planet Gearshaft	RGB
		Rings, AGB Coupling Shaft Brg	AGB
		Rings, FCU Drive Brg	AGB
		Rings, Start-Gen Drive Brg	AGB
		Rings, Hydraulic Pump Drive Brg	AGB
		Rings, Vacuum Pump Drive Brg	AGB
		Rings, External Scavenge Pump Drive Brg	AGB
AMS 6490	Steel (M50)	No. 2 Brg, Rings and Rollers	GG
		No. 3 Brg Rings and Rollers	PT
AMS 6491	Steel (M50)	No. 1 Brg Rings and Balls,	GG
		PT Shaft	PT
		PT Shaft Coupling	PT
		No. 4 Brg Rings and Balls,	PT
AMS6265/ 6444	Steel	Shaft, Second-stage Planet Gear	RGB
		Shaft, First-stage Planet Gear	RGB
AMS 4928	Titanium Alloy	Compressor, First-stage	GG
		Stubshaft, First-stage Compressor	GG

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Specification	Material	Nomenclature	Engine Location
	Molybdenum	Coating, No. 2 Brg, Labyrinth, Seals, Air	GG

Table 206 Common Contaminants - Non-organic Material

Contaminant Detected	Main Elements	Cause of Contaminants
Alumina	Al_2O_3	Manufacturing
Black Magnetic	Fe_3O_4	Anaerobic decomposition of shaft material, can look like carbon is a corrosive product.
Calcium Oxide	CaO	Environment
Rust, Non-magnetic	Fe_2O_3	Rust
Sand	SiO_2	Manufacturing/Environment
Silicon Carbide (Black, Shiny, Angular)	SiC	Manufacturing/Blasting
Zirconia	$ZrSiO_4$	Manufacturing
	$Mo5_2$	Common as a trace, could combine with Fe (Iron) to indicate bearing alloy. Usually found in new or overhauled engines.
Calcium Chloride (Salt)	$CaCl_2$	Environment
Sodium Chloride (Salt)	NaCl	Environment
Calcium Sulphate (Salt)	$CaSO_4$	Environment

Table 207 Common Contaminants - Organic Material

Contaminant Detected	Source
Black Fluorocarbon Rubber, Red Silicon Rubber, Blue Fluorosilicon Rubber	Preformed Packings, Seal Gaskets
Chloroprene Nitrile Rubber	Gaskets
Carbon	Decomposed Oil, Carbon Seals
Fibers	

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Contaminant Detected	Source
Plastics	
Paint Flakes	

F. Laboratories

(1) Chip and flake analysis must be done by approved laboratories; refer to Service Information Letter No. GEN-123 for the list of approved oil laboratories.

10. Marking of Parts

A. General

(1) Marking of engine parts, assemblies, or weldments shall be applied so as to ensure maximum legibility and durability of mark but in a manner that will not affect function or serviceability of part. Only applicable Pratt & Whitney Canada marking methods shall be used.

(2) Except where otherwise specified, reidentification of parts shall be accomplished adjacent to, or in a location similar to that of, original marking.

(3) All marking characters, unless otherwise specified, shall be 0.060 to 0.160 inch high. In special cases, when marking area is governed by size or configuration of part, characters not less than 0.016 inch nor more than 0.200 inch in height are permitted.

(4) Electric-arc scribing, particularly hand arc scribing, whereby characters are produced by the action of an electric arc between the surface and an the electrode (scriber), has been found unsuitable for jet engine parts and must not be used.

(5) Acid etching, whereby characters are formed by action of an acid on the surface of the part, is not recommended because of its possible corrosive effect.

(6) Do not use Soapstone to mark the engine parts.

B. Permanent Marking

(1) General

(a) Marking should be done as close to original marking as possible. Permanent methods of marking are those in which marking is legible during entire service life of part.

(b) Permanent markings shall not extend onto any radius, chamfer, sharp edge, or fillet adjoining designated marking surface.

(2) Approved Marking Methods

CAUTION: DO NOT USE ELECTROLYTIC ETCH METHOD OF MARKING ON ANODIZED ALUMINUM SURFACES.

(a) Electrolytic Etch - Characters are produced by electrolysis confined to area of characters by a stencil. This technique is sometimes used as a temporary method. Do not confuse it with electric-arc scribing.

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(b) Vibration Peening - Characters are produced by a vibrating, radius-tipped, conical tool.

Manual - tool is hand-guided and has a single tip.

Mechanical - tool is mechanically guided and has a single tip, or has multiple tips producing one or more complete characters simultaneously. This method is not usually permitted if hardness of part at time of marking is higher than Rockwell C45 or equivalent.

NOTE: Vibration peening marking method can be substituted for marking parts originally marked by diamond drag or roll marking.

C. Temporary Marking

(1) General

(a) Temporary methods of marking are those in which the marking will identify the parts during ordinary handling, and storage, and final assembly.

CAUTION: LEAD AND/OR METALLIC PENCILS, OR ANY TEMPORARY MARKING METHOD LEAVING A DETRIMENTAL DEPOSIT OF CARBON, ZINC, COPPER, LEAD, OR SIMILAR RESIDUE WHICH MAY CAUSE A REDUCTION IN FATIGUE STRENGTH AS A RESULT OF CARBONIZATION OR INTERGRANULAR ATTACK WHEN THE PART IS SUBJECTED TO INTENSE HEAT, SHALL NOT BE USED. THE MARK MUST BE THOROUGHLY REMOVED BY A PROCEDURE DICTATED BY THE MARKING METHOD AND THE PART MATERIAL.

(b) When using a marking pencil, marks shall not be applied to carbon seal plate surfaces or to mating surfaces of finished machined parts. Heavy deposits of marking material could adversely affect clearance and runout.

(2) Approved Marking Methods

CAUTION: DO NOT USE ELECTROLYTIC ETCH METHOD OF MARKING ON ANODIZED ALUMINUM SURFACES.

(a) Electrolytic Etch - Characters are produced by electrolysis confined to area of the characters by a stencil. This is primarily a permanent marking method. However, where specified, it may also be used for temporary marking.

(b) Ink Marking

1 An ink that does not damage the surface is used to produce the characters. Inks used in marking may have a light etching action, but does not damage the surface.

NOTE: Ordinarily, ink stamping and electrolytic etching, when used as a temporary marking method, may be applied to any surface which, after assembly, does not move relative to a contacting surface.

(3) Marking Materials, Hot and Cold Section Engine Parts

(a) Felt Wick Pen (PWC05-048A) and Instant Dry Ink (PWC05-046).

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(b) Marks-A-Lot Marker(PWC05-048B)

(c) Brushpen No. 57 (PWC05-048) and Fast Drying Ink (PWC05-046A), black, red and purple.

(d) Micro Supreme Dye, No. 142 (Purple) (PWC05-002).

(e) Berol Verithin Pencil, Red No. 750 (PWC05-018F). May be used on parts which are not directly exposed to the gas path. This type of mark is easily obliterated and, therefore, less durable.

(f) Design Spectracolor Pencil, Silver No. 1428 (PWC05-018), or Verithin Pencil, Silver No. 753 (PWC05-018A).

(4) Marking Materials, Cold Section Engine Parts Only

Dykem Ink (PWC05-046C)KX425 (Black) or (PWC05-332)KXX122 (White).

Carters Ink (PWC05-027) KX451 (Black).

(5) Marking Materials, Hot Section Engine Parts Only. Use layout dye (PWC05-041) (lightly applied) to mark parts which are directly exposed to engine gas path such as turbine blades and disks, turbine vanes, and combustion chamber liner.

11. Lubrication of Preformed Packings

A. General

CAUTION: APPLICATION OF LUBRICANT PLUS PROPER ASSEMBLY, WILL PREVENT DAMAGE TO PREFORMED PACKINGS. DAMAGED PREFORMED PACKINGS CAN CAUSE ENGINE MALFUNCTION.

(1) Prior to installation, it is recommended that all preformed packings be coated with lubricant to the following specifications:

(a) Preformed packings used in the fuel and/or oil system, use engine oil (PWC03-001).

12. Touch-up Solution

WARNING: REFER TO THE MATERIAL SAFETY DATA SHEETS BEFORE YOU USE THESE MATERIALS FOR INFORMATION SUCH AS HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES. SOME OF THESE MATERIALS CAN BE DANGEROUS. YOU CAN GET THE DATA SHEETS FROM THE MANUFACTURERS OR THE SUPPLIERS OF THESE MATERIALS.

A. Chrome Pickle Solution (PWC05-161) (Magnesium RGB and AGB Housings) (Ref. Table 208)

(1) Fill a suitable container with tap water to no more than $^{1}/_{2}$ of the volume of the solution required.

(2) Slowly and carefully add the necessary quantity of sodium dichromate (PWC05-055). Mix to dissolve the sodium dichromate.

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(3) Slowly and carefully add the necessary quantity of nitric acid (PWC05-195). Mix the solution.

(4) Fill the remainder of the container with tap water to the specified volume required.

Table 208 Chrome Pickle Solution

MATERIAL	PROPORTION	PROPORTIONAL LIMITS
Sodium Dichromate (PWC05-055)	**1.5 lbs.**	**1.0 to 1.5 lbs.**
Nitric Acid (PWC05-195)	1.5 pints	1.0 to 1.5 pints
NOTE: The figures that are given are per gallon of solution required.		

B. Chromate Repair of Magnesium Casings - Method A

WARNING: REFER TO THE MANUFACTURER'S MATERIAL SAFETY DATA SHEETS FOR CONSUMABLE MATERIALS INFORMATION. IT GIVES INFORMATION ABOUT HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL PROPERTIES, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES.

(1) Prepare chromate conversion solution (PWC11-019) (Ref. Table 209):

(a) Fill an applicable container to 3/4 level with tap water.

(b) Slowly and carefully add necessary quantity of chromate conversion salts (PWC05-198).

(c) Slowly and carefully add necessary quantity of hydrochloric acid (PWC05-196). Mix the solution.

(d) Slowly and carefully add necessary quantity of wetting agent (PWC05-162). Mix the solution.

(e) Fill the container to operating level. Mix the solution.

(f) Add hydrochloric acid (PWC05-196) or a combination of sodium hydroxide solution (PWC05-057) and 16 oz of distilled water (PWC05-073) to adjust pH as necessary. Mix the solution.

Table 209 Chromate Conversion Solution (PWC11-019)

Material	Make-up	Operating Limits
Chromate conversion salts - (PWC05-198)	**5 oz. wt/gal**	**4.5 - 5.5 oz. wt/gal**
Hydrochloric acid (PWC05-196)	2 fl oz/gal	1 - 3 fl oz/gal
Wetting agent	1 ml/gal	

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Material	Make-up	Operating Limits
Chromate conversion salts - (PWC05-198) (PWC05-162)	**5 oz. wt/gal**	**4.5 - 5.5 oz. wt/gal**
Sodium hydroxide solution (PWC05-196) and 16 oz distilled water (PWC05-073)	Add as necessary to increase pH to operating range.	pH = 0.6 - 1.1
NOTE: To maintain wetting agent (PWC05-162), add 0.5 ml for each 5 oz of chromate conversion salts (PWC05-198) added.		

C. Chromate Repair of Magnesium Casings - Method B

(1) Put an applicable cover or debris collector to the necessary area to prevent the entry of debris into the compressor or into other openings. Attach a lanyard to the debris collector or cover so it will not fall into the compressor cavity.

(2) Clean the surface of the magnesium casing with one of the methods that follow:

- A swab soaked in isopropyl alcohol (PWC11-014B)
- A coated abrasive crocus cloth (PWC05-061)
- An isopropyl alcohol hand-wipe or other applicable hand-wipe chemical

(3) Remove the corrosion with one of the materials that follow:

- A small non-metallic wire brush
- A stainless steel or aluminum brush
- A coated abrasive crocus cloth (PWC05-061)
- An abrasive cloth with aluminum oxide or glass grit. Do not use silicon carbide abrasive.
- For better results, you can use a swab soaked in a chromic acid solution (PWC05-197) that has a temperature of 122.5 ± 40.5 °C (252.5 ± 72.5 °F).

(4) Flush with clean water at room temperature.

(5) Flush with clean water at a temperature of 60 °C (140 °F) minimum.

(6) Dry with clean low pressure compressed air at 200 kPa (29 psig).

(7) Apply chromate conversion solution (PWC11-019) or chrome pickle solution (PWC05-161) at a temperature of 23 ± 6 °C (74 ± 11 °F) for 30 to 45 seconds to the prepared surface with a swab or brush. Apply again frequently to make sure that the surface is wet with solution during the treatment.

(8) You can use the materials that follow as an alternative to chromate conversion solution (PWC11-019) or chrome pickle solution (PWC05-161):

(a) Touch-up conversion coating (PWC05-418) - pen applicator for non-hexavalent

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chromium based solution.

(b) Touch-up conversion coating applicator (Alodine) (Magnesium Treatment Kit) (PWC05-419).

(9) Swab area with clean water until swabs are no longer stained yellow.

(10) Dry the part in an oven or locally with a heat gun for 20 to 30 minutes at 131 ± 46 °C (267.5 ± 82.5 °F).

D. Repair of Protective Coating on Painted Surfaces

NOTE: The instructions that follow are not applicable for flanges, engine mount pads or accessories mount pads.

(1) Touch-up exposed metal (Ref. Para. 12.A., B. and C.).

(2) Mask the area where painting is not necessary.

(3) Apply two layers of epoxy primer (PWC13-001). Wait one hour before you apply second layer. Let the part dry at room temperature for 8 hours minimum after the second layer.

(4) Apply three layers of aluminized epoxy enamel (PWC05-037). Wait 15 to 30 minutes before you apply next layer. Let the last coat dry at room temperature for 24 hours minimum.

(5) Remove the masking. Make sure that the exposed area is fully covered.

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